De La Salle University

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Risk-Off Shocks and Spillovers in Safe Havens

Replication and Extension

In partial fulfillment of the requirements for

FINARTS Section C01

Department of Financial Management

Term 3, Academic Year 2025–2026

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Date Submitted on

7 August 2026

Abstract

This study replicates Beirne and Sugandi (2023) for Japan and extends the sample through June 2026. A structural vector autoregression with Cholesky identification and a block-exogenous risk-off equation is estimated at the daily and monthly frequencies over the paper period with 2021-era data vintages, and again over the extended sample with current-vintage data, under a dual-vintage policy that separates the replication from the extension. Nine of the seventeen variable-frequency impulse response comparisons against the published figures are matches, five are partial matches, and three are mismatches. The paper's core results reproduce over the original sample. The yen appreciates in real effective terms after a risk-off shock, the JGB spread widens, real GDP declines with a lag, and portfolio flows do not respond significantly. The mismatches are confined to the uncertainty index at both frequencies and to other investment at the daily frequency. The extension shows the output channel weakening, with the monthly real GDP response statistically indistinguishable from zero after 2021 while the price channels persist. The restricted model's output response flips sign when current-vintage data replace the 2021-era series, which makes data vintage a first-order replication choice. Direct investment is the only capital flow with a statistically significant response, at the three-month horizon. The findings imply that the yen's safe-haven mechanism has become conditional on the monetary policy constellation and that a faithful replication of the paper requires the data as the original authors observed them in 2021.

Keywords: safe haven currencies, risk-off episodes, structural VAR, Japan, yen, capital flows

Contents

1 Introduction

Global investors rebalance toward assets that preserve value when financial markets turn turbulent. Baur and Lucey (2010) define a hedge as a security that is uncorrelated with stocks or bonds on average, and a safe haven as a security that is uncorrelated with stocks and bonds during a market crash. Hossfeld and MacDonald (2014) transpose the same distinction onto currencies, classifying a safe haven currency as one whose effective returns are negatively related to global stock market returns in periods of high financial stress. The Japanese yen and Japanese government bonds (JGBs) are the archetypal case. Foreign investors bought Japanese assets during the 2001 dot-com collapse and the 2007 to 2008 global financial crisis, and the yen has appreciated against the dollar in nearly every major risk-off episode since the 1990s (Beirne & Sugandi, 2023). The stakes of this behavior are real. Safe-haven inflows appreciate the currency, erode export competitiveness, and can transmit a global shock into the domestic economy through the exchange rate.

Beirne and Sugandi (2023) provide the most complete accounting of these spillovers for Japan. The replicated article appears in the Pacific-Basin Finance Journal, which carries an A classification on the Australian Business Deans Council Journal Quality List. Their study estimates country-specific structural vector autoregressions (VARs) for Japan, Switzerland, and the United States on working-days data and identifies four results for Japan. The yen's real effective exchange rate (REER) appreciates sharply and persistently after a risk-off shock. Portfolio flows to Japan show no significant response, which the authors attribute to rapid adjustment of financial markets toward new equilibrium exchange rates. Negative real spillovers appear only for Japan, with the appreciation amplifying the decline in GDP growth. Domestic economic policy uncertainty does not respond significantly. The surrounding literature is consistent with these patterns. Habib and Stracca (2012) identify net foreign asset positions as the most robust predictor of safe haven status. Ranaldo and Söderlind (2010) and Fatum and Yamamoto (2015) document the yen's crisis-time appreciation, and Botman et al. (2013) show that yen appreciation during risk-off episodes coincides with offshore derivative trading rather than measurable cross-border flows.

Three developments since the paper's sample ended in March 2021 limit what the published evidence can say today. The first is the regime shift in the yen itself. Between 2021 and 2026 the currency depreciated from roughly 105 to 160 against the dollar despite foreign exchange interventions in 2022 and 2024 that arrested but did not reverse the slide, while the Bank of Japan exited yield curve control and the Federal Reserve raised rates aggressively, an environment in which the traditional safe-haven response weakened. The second is data revision. Japan's real GDP underwent a base-year change and subsequent benchmark revisions that raised current-vintage output by 3.5 to 6.3 percent over 2019 to 2021, and the BIS retroactively rebased its REER from a 2010 base to a 2020 base, so a researcher who downloads these series today does not obtain the data the original authors observed. The third is the set of unreported estimation settings. The paper does not state which quadratic interpolation variant was used, what lag maximum was searched, or how the confidence intervals were computed, and any independent replication therefore carries uncertainty about whether differences originate in data, code, or configuration.

This study replicates Beirne and Sugandi (2023) for Japan and extends the sample through June 2026. The replication rebuilds the paper's recursively restricted VAR with a Cholesky identification scheme, adds the unrestricted specification as a robustness check, and estimates both at the daily and monthly frequencies. The data follow a dual-vintage policy. The paper period from 14 January 1999 to 31 March 2021 uses the 2021-era vintages of real GDP and the REER, and the extension from April 2021 to 30 June 2026 uses current-vintage data. The daily financial market series were downloaded from LSEG Workspace, and the macroeconomic series were collected via automated Python API scripts from the Federal Reserve Bank of St. Louis ALFRED archive, the BIS, the Bank of Japan, and the World Uncertainty Index database.

Beirne and Sugandi (2023) set out to quantify real output spillovers, real effective exchange rate appreciation, sovereign yield spread widening, portfolio capital flow neutrality, and domestic policy uncertainty in Japan during global risk-off episodes. To independently evaluate whether these empirical objectives reproduce under an open-source framework and survive into the post-2021 monetary policy divergence regime, the primary objective of this study is

To evaluate the empirical validity, temporal generalizability, and data-vintage sensitivity of risk-off spillover dynamics in Japan by independently replicating and extending the structural vector autoregression framework of Beirne and Sugandi (2023).

To achieve this overarching aim, the research addresses three central research questions

- *Baseline Qualitative Structure*

  Does the paper's qualitative structure for Japan reproduce when the structural vector autoregression is re-estimated independently on the original sample period?

- *Post-2021 Regime Generalizability*

  Do the estimated macroeconomic and financial relationships survive into the post-2021 regime of monetary policy divergence and yen depreciation?

- *Data Vintage Sensitivity*

  Which empirical findings remain robust to historical data revisions, and which results depend strictly on the historical data vintage observed by the original authors?

To answer these research questions, the study pursues three specific secondary objectives

- *Rebuild Baseline SVAR Models*

  To rebuild the restricted and unrestricted structural VAR models for Japan over the original 1999 to 2021 sample using period-appropriate data vintages and compare the estimated impulse responses directly against published benchmarks.

- *Evaluate Out-of-Sample Generalizability*

To re-estimate both econometric specifications over the extended sample through June 2026 using current-vintage data and document the resulting regime shifts in exchange rate and output spillovers.

- *Disclose Specification and Calibration Choices*

  To disclose every specification choice, model calibration procedure, and software implementation detail so that the boundary between empirical data effects and configuration choices is completely explicit.

The study contributes in three ways. It validates the paper's core results for Japan, since the daily restricted VAR reproduces the negative real GDP response, the REER appreciation, the bond spread widening, and the null capital flow responses. It extends the evidence into a regime in which the yen's safe-haven status is disputed, where the real GDP response weakens toward zero. And it demonstrates a data-vintage problem that is material to replication, since the restricted model's GDP response flips sign when current-vintage data replace the 2021-era series.

The remainder of the paper is organized as follows. Section 2 reviews the literature on safe haven currencies and develops the theoretical framework and hypotheses. Section 3 describes the data and the econometric methodology. Section 4 presents the descriptive statistics, stylized facts, and impulse response results. Section 5 discusses the findings in relation to theory and the original paper, and Section 6 concludes. The literature review that follows assembles the currency evidence and derives the five hypotheses that the methodology in Section 3 is designed to test.

2 Review of Related Literature and Theoretical Framework

2.1 Safe Haven Currencies

The currency literature builds on the hedge and safe haven definitions set out in the introduction. The safe haven concept in its modern form begins with the distinction between hedging and safety. Baur and Lucey (2010) show for gold that an asset can be uncorrelated with equities and bonds on average without providing protection in a crash, and their hedge-safe haven dichotomy has become the standard vocabulary. Hossfeld and MacDonald (2014) apply the distinction to exchange rates and find that the Swiss franc and the US dollar qualify as safe haven currencies among the G10, while yen appreciation in crises is driven mainly by the unwinding of carry trades rather than by safe haven inflows. The empirical record on the yen is otherwise strong. Ranaldo and Söderlind (2010) find that the Swiss franc and the yen appreciate against the dollar and the euro when global equity markets fall and volatility rises. Fatum and Yamamoto (2015) go further, ranking the yen as the safest of the safe haven currencies because it appreciates as market uncertainty increases regardless of the prevailing level of uncertainty, whereas other currencies react nonlinearly. Cho and Han (2021) confirm in a cross-quantilogram framework that the yen, the euro, and the Swiss franc appreciate in periods of high foreign exchange volatility and rising US Treasury yields.

The literature also identifies the structural conditions that make a currency a safe haven. Habib and Stracca (2012) stress net foreign asset positions, which survive as the most consistent predictor of safe haven status after controlling for carry trade, along with financial development and foreign exchange market liquidity. Low interest rates create the carry trade that positions the yen as a funding currency, and Nishigaki (2007) shows that US stock prices dominate the behavior of yen carry positions while the interest rate differential with Japan plays a smaller role. Brunnermeier et al. (2009) demonstrate that carry trade returns crash when funding currencies appreciate, which is precisely the mechanism that produces yen strength during risk-off episodes.

Imakubo et al. (2015) formalize the demand side through a currency premium that depends on deviations of real interest rates and real exchange rates from equilibrium.

2.2 The Yen as a Safe Haven Currency

Japan-specific evidence refines the general picture in one important respect. Botman et al. (2013) conduct a forensic analysis of yen appreciation during risk-off episodes and find that the appreciation is unrelated to capital inflows and to expectations about relative monetary policy. The appreciation coincides instead with portfolio rebalancing through offshore derivative transactions, where reduced net short positions and a buildup of net long positions in the yen move the spot rate without appearing in the balance of payments. This finding matters for the present study because it predicts a null response of measured capital flows alongside a strong exchange rate response, exactly the pattern that Beirne and Sugandi (2023) report.

The real economy consequences of safe haven status are well documented for Japan. Masujima (2017) argues that yen strength driven by safe haven demand slows the post-crisis recovery through net exports, and that large-scale monetary easing in response may mask financial vulnerabilities related to public debt sustainability. Belke and Volz (2020) trace the appreciation to the structural hollowing out of Japanese industry. Iwaisako and Nakata (2017) caution against overstating the export channel, noting that aggregate global demand played the larger role in the trade decline after the 2008 crisis, but their structural VAR still finds economically meaningful exchange rate effects on exports. On the bond side, Lam and Tokuoka (2013) attribute the low and stable JGB yields to steady domestic inflows, high domestic ownership, and safe haven demand, while warning that population aging and the decline of private-sector savings threaten this stability.

2.3 Real Effects and Financial Vulnerabilities

Beyond the Japan-specific evidence reviewed above, the general literature on safe assets connects safe haven demand to the real economy through equilibrium interest rates and adjustment costs. Habib et al. (2020) show that high demand for safe assets in crisis times depresses the natural real interest rate, with the exchange rate appreciation and the contraction in global demand disrupting normal adjustment mechanisms. Bussière et al. (2013) document that large real appreciations create adjustment costs that can lead to economic dislocation when exchange rates eventually revert. For economies with low inflation and policy rates near the zero bound, real appreciations driven by risk-off episodes can feed deflationary pressures and weigh on aggregate demand. These mechanisms give the negative output response to risk-off shocks in safe haven destinations a clear theoretical basis.

Uncertainty expectations interact with the exchange rate through a separate channel. Beckmann and Czudaj (2016) find that exchange rate expectation errors decrease when uncertainty rises, particularly for monetary policy uncertainty, and that the yen is exceptional in appreciating when uncertainty increases. This result is relevant to the uncertainty variable in the empirical model, because it suggests that the yen's response to risk-off shocks operates through a repricing of risk rather than through a measurable rise in domestic economic policy uncertainty, which would reconcile a muted uncertainty response with a strong currency response.

2.4 Risk-Off Episodes and Capital Flows

The mechanisms described above are triggered by identifiable episodes of market stress, and the empirical definition of a risk-off episode used throughout this literature follows De Bock and de Carvalho Filho (2013), who identify a risk-off event as a day on which the Chicago Board Options Exchange Volatility Index (VIX) exceeds its 60-day backward-looking moving average by 10 percentage points. Their study of currency behavior during such episodes finds that high-yield currencies depreciate and low-yield currencies appreciate, consistent with carry trade unwinding, and that the yen behaves as the clearest case of a low-yield currency gaining in stress periods.

Capital flow dynamics in response to risk-off shocks are governed by portfolio rebalancing. Hau and Rey (2006) document a strong negative comovement between exchange rates and relative equity market returns, driven by repatriation and portfolio rebalancing flows. When global investors sell foreign equities and repatriate funds, the funding currency appreciates even without a change in measured portfolio inflows. This repatriation mechanism, together with the offshore derivative channel of Botman et al. (2013), explains why capital flows measured in the balance of payments can be unresponsive to risk-off shocks while the exchange rate moves sharply.

2.5 The Post-2021 Regime and the Yen's Safe Haven Status

The safe haven evidence summarized above is built almost entirely on data through 2021, and the years since then have called the yen's status into question. Beirne and Sugandi (2023) themselves flag the onset of the break. In their footnote on the period following the Russian invasion of Ukraine in February 2022, they note that widening yield differentials between the United States and Japan, rising energy prices, and the worsening of Japan's balance of payments position drove a sharp yen depreciation against the dollar, with the dollar soaring against traditional safe haven currencies including the yen and the Swiss franc. The Federal Reserve's aggressive tightening cycle, which took the policy rate to 5.25 to 5.5 percent against a near-zero Bank of Japan policy rate, pushed the dollar-yen rate to 145 by September 2022 and past 150 by October, its weakest level since 1990. The Ministry of Finance intervened in September and October 2022, its first intervention since 1998, spending roughly 9.2 trillion yen to arrest the slide, and the currency then traded between 130 and 152 through 2023 as the Bank of Japan twice adjusted its yield curve control framework. The push to 160 in April 2024 drew a second intervention round of roughly 9.8 trillion yen, the first since October 2022, which capped the depreciation near that level. The yen's crisis appreciation became conditional on the rate differential rather than automatic.

Recent empirical work formalizes this conditionality. Park and Fang (2025) estimate time-varying intra-safe haven currency behavior among the dollar, the franc, and the yen and find that the hierarchy among safe haven currencies shifts over time, with the yen's crisis response depending on the type of shock and the prevailing monetary policy constellation. Lu et al. (2024) reach the same conclusion from the East Asian market, showing that no East Asian currency, including the yen, qualifies as a safe haven when uncertainty is measured by geopolitical risk or trade policy uncertainty, while the yen retains safe haven properties under the conventional VIX-based risk measure. The International Monetary Fund's surveillance attributes the yen's persistent weakness to the interest rate differential with the United States and documents a 2024 depreciation of roughly 8 percent with exchange rate volatility beyond what rate expectations alone explain (International Monetary Fund, 2024, 2025). This stands in contrast to the unconditional ranking of Fatum and Yamamoto (2015), who found the yen to be the safest of the safe haven currencies over the global financial crisis period. The difference between the two studies is not a contradiction. It is evidence that the safe haven mechanism itself is regime-dependent, which is precisely the property that a sample extension can test.

The August 2024 episode sharpens the point. A Bank of Japan rate hike combined with a weak United States jobs report triggered a rapid unwind of leveraged yen carry positions in early August 2024, with the TOPIX index falling 12.4 percent on 5 August and the VIX surging above 65, a risk-off rally in the yen driven substantially by carry trade deleveraging rather than a vote of confidence in the Japanese economy (Aquilina et al., 2024; Bank for International Settlements, 2024; International Monetary Fund, 2025). The yen appreciated sharply during this episode, but the appreciation was the mechanics of carry trade deleveraging rather than a fresh wave of safe haven inflows. The episode sits inside the extension window of the present study and appears in the risk-off episode table in Section 4.

Domestic policy developments complete the picture. The Bank of Japan ended yield curve control and exited its negative interest rate policy in March 2024, its first policy rate increase since 2007, and raised the policy rate in five steps to 1.00 percent by June 2026, its highest level in three decades, with 10-year JGB yields rising above 2.5 percent after more than a decade near zero, and the Nikkei 225 more than doubled between 2023 and 2026 on the back of corporate governance reforms and the return of inflation. The April 2025 tariff shock provided the clearest counterexample to the depreciation narrative, with the yen strengthening toward 140 as the dollar failed to behave as a safe haven, through carry trade unwinding and haven flows, before the currency weakened again toward 160 by mid-2026 as the rate differential reasserted itself. The post-2021 sample therefore contains a different monetary policy constellation, a different external position, and a different equity market regime. Whether the structural relationships estimated over the paper period survive in this environment is an open empirical question, and it is the question that the extended sample in this study is designed to answer.

2.6 Theoretical Framework

The theoretical framework synthesizes the evidence reviewed above into three transmission channels from a global risk-off shock to the Japanese economy. The first is flight to safety, through which a rise in global risk aversion increases demand for yen-denominated assets, appreciating the yen in real effective terms and compressing JGB yields relative to US Treasury yields. The second is carry trade unwinding, through which the yen, as the traditional funding currency, is bought back by leveraged investors when a risk-off shock hits, amplifying the appreciation beyond what safe haven demand alone would produce. The third is portfolio rebalancing and offshore derivative activity, through which investors adjust yen positions in derivatives and repatriation flows that move the exchange rate without registering as portfolio inflows in the balance of payments, separating the price response from the volume response.

The framework yields directional predictions for the endogenous variables in the model. A risk-off shock should appreciate the yen REER, widen the JGB yield spread relative to the United States, and depress real GDP through the competitiveness channel with a lag of one to two quarters. Capital flow variables should show no systematic response, because the price channels adjust through offshore activity. Domestic economic policy uncertainty should be muted relative to other safe havens, because market participants price in the repatriation of overseas holdings. The framework does not deliver an unambiguous prediction for the stock market, where the surge-to-safety effect on Japanese equities competes with the negative wealth and exporter effects of appreciation.

2.7 Statement of Hypotheses

To evaluate the five transmission channels established in the theoretical framework, five formal hypothesis pairs are formulated. Each hypothesis pair consists of a Null Hypothesis (H_0) of zero effect and an Alternative Hypothesis (H_1) derived from international macroeconomic theory

- *Hypothesis 1 (Exchange Rate Appreciation Channel)*

  $H_{0,1}$: A global risk-off shock has no statistically significant effect on Japan's real effective exchange rate.

  $H_{1,1}$: A global risk-off shock induces a statistically significant and persistent real effective appreciation of the Japanese yen.

- *Hypothesis 2 (Real Output Contraction Channel)*

  $H_{0,2}$: A global risk-off shock has no statistically significant effect on Japan's real gross domestic product.

  $H_{1,2}$: A global risk-off shock induces a statistically significant decline in Japan's real gross domestic product through the real exchange rate competitiveness channel with a lag.

- *Hypothesis 3 (Sovereign Yield Spread Widening Channel)*

  $H_{0,3}$: A global risk-off shock has no statistically significant effect on the yield spread between 10-year Japanese Government Bonds and 10-year US Treasury notes.

  $H_{1,3}$: A global risk-off shock induces a statistically significant widening of the yield spread as safe-haven demand compresses Japanese yields relative to US yields.

- *Hypothesis 4 (Capital Flow Neutrality Channel)*

  $H_{0,4}$: A global risk-off shock has no statistically significant effect on net portfolio investment inflows to Japan.

$H_{1,4}$: A global risk-off shock induces a statistically significant change in net portfolio investment inflows to Japan.

- *Hypothesis 5 (Domestic Policy Uncertainty Channel)*

  $H_{0,5}$: A global risk-off shock has no statistically significant effect on Japan's domestic economic policy uncertainty index.

  $H_{1,5}$: A global risk-off shock induces a statistically significant increase in Japan's domestic economic policy uncertainty index.

Section 3 details the two-tier structural vector autoregression data pipeline and econometric methodology designed to test these five hypotheses.

3 Methodology

3.1 Research Design

The design set out in this section tests the five hypotheses of Section 2. This study follows the empirical strategy of Beirne and Sugandi (2023, Section 4) and modifies it in documented ways. The object of estimation is a country-specific structural VAR for Japan on working-days data, organized in two tiers. Tier one is the daily restricted VAR over the paper period, the specification that produces the paper's published headline figures. Tier two is the monthly VAR over the paper period and the extended sample, which matches the paper's appendix structure and provides the frequency robustness check. The paper period runs from 14 January 1999 to 31 March 2021, and the extended sample runs through 30 June 2026.

The scope of the replication is the Japanese model. The paper estimates its VARs for three safe haven destinations, and this study re-estimates the Japanese model only, with the Swiss and United States estimates entering the discussion only as the published evidence reports them. The paper's weekly frequency panels are likewise not replicated, and the two-tier design covers the daily and monthly frequencies only. Both exclusions are scope decisions, disclosed so that the boundary of the replication claim is explicit.

The working-days calendar is constructed by removing weekends from the full daily calendar, keeping only Monday through Friday. The sample start is the paper's first working day on which all financial series are available. The daily estimation sample contains 5,199 complete-case observations in the paper period and 1,234 in the extension, for a full-sample total of 6,433 working days. The monthly tier contains 264 complete months for the paper period and 63 for the extension. The risk-off variable is anchored to the VIX, which trades on every United States market day, and a left join preserves dates on which the VIX exists but the Japanese series do not, such as Japanese holidays, so that risk-off episode start dates falling on Japanese non-trading days are retained.

3.2 Data and Sources

Two groups of sources supply the data for the design. The daily financial market series were downloaded from LSEG Workspace. These are the VIX, the Nikkei 225 index, the bilateral USD/JPY exchange rate, the 10-year JGB yield, and the 10-year US Treasury yield. All LSEG exports share a six-column format with the date and open, high, low, close, and volume fields, so a single parser handles them. The JGB file is the exception in layout, with a title row that is skipped and the 10-year yield stored in the eleventh column, and rows without a valid 10-year value are dropped. The spread variable is computed as the 10-year JGB yield minus the 10-year US Treasury yield.

The macroeconomic series were collected via automated Python API scripts from their official sources. Real GDP and nominal GDP come from the Federal Reserve Bank of St. Louis, FRED series JPNRGDPEXP and JPNNGDP, retrieved in 2026. The World Uncertainty Index (WUI) for Japan comes from the World Uncertainty Index database maintained by Ahir, Bloom, and Furceri, retrieved through FRED as series WUIJPN at the quarterly frequency. The real effective exchange rate comes from the BIS broad index. The capital flow series come from the Bank of Japan's BPM6 balance of payments statistics, measuring the net incurrence of portfolio investment liabilities, other investment liabilities, and direct investment liabilities.

The capital flow extraction is the most demanding part of the data preparation. Pre-2014 observations come from the Bank of Japan's 6pi-1 portfolio summary, a Shift-JIS encoded CSV with a mixed layout that requires reading the raw lines, locating the monthly figures section, and parsing a 29-column table in which the equity, long-term debt, and short-term debt net liability columns sit at fixed positions. The values are comma-formatted and use a dash placeholder for missing observations, and the dates are written in Japanese era format, which is converted to the Gregorian calendar with the Heisei era offset of 1988 and the Reiwa offset of 2018. Post-2014 observations come from the Bank of Japan's BPM6 API files, which are simple two-column CSVs in units of 100 million yen. The debt securities series is the sum of the long-term and short-term debt securities components, consistent with the Bank of Japan's D-04 reporting convention, and the pre-2014 and post-2014 segments are concatenated into continuous series for each flow category.

Table 1 defines each variable, its frequency, and its source. The vintage arrangements for the paper period are described in the next subsection.

Table 1: Endogenous Variables, Definitions, and Sources

Variable	Definition	Frequency	Source
Risk-off	Indicator equal to 1 when the VIX exceeds its 60-day moving average by at least 10 points	Daily	LSEG Workspace, authors' calculation
Log (WUI)	Log of the World Uncertainty Index for Japan	Quarterly to daily	World Uncertainty Index database
Spread	10-year JGB yield minus 10-year US Treasury yield	Daily	LSEG Workspace
Log (RGDP)	Log of constant-price real GDP	Quarterly to daily	FRED ALFRED (vintage 2021-06-01)
Log (REER)	Log of the real effective exchange rate, broad index	Monthly to daily	BIS (Wayback 2021-05-24)
Log (Nikkei)	Log of the Nikkei 225 index	Daily	LSEG Workspace
Debtsec	Net portfolio investment inflows to debt securities, percent of nominal GDP	Monthly to daily	Bank of Japan
Equity	Net portfolio investment inflows to equity, percent of nominal GDP	Monthly to daily	Bank of Japan
Other	Net other investment inflows, percent of nominal GDP	Monthly to daily	Bank of Japan
Direct	Net direct investment inflows, percent of nominal GDP	Monthly to daily	Bank of Japan

Note. Capital flow variables follow BPM6 conventions. Positive values indicate capital inflows. The nominal GDP denominator is current-vintage throughout.

3.3 Dual-Vintage Data Policy

The data follow a dual-vintage policy because the published sample has been retroactively revised since 2021. Japan's real GDP underwent a base-year change from chained 2011 yen to chained 2015 yen in December 2020, and subsequent benchmark revisions raised current-vintage output by 3.5 to 6.3 percent over 2019 to 2021 while flattening the COVID-19 trough. The BIS rebased its REER from 2010 equal to 100 to 2020 equal to 100 in January 2023, revising the entire historical series and trade weights.

The paper period therefore uses the real GDP vintage of 1 June 2021 from the FRED ALFRED archive, retrieved through the FRED API with the vintage date parameter set to 2021-06-01. The REER for the paper period uses the BIS broad index snapshot of 24 May 2021, recovered through the Internet Archive Wayback Machine from the archived BIS file, because the BIS does not publicly archive retrospective vintages and the Wayback snapshot is the closest approximation of the file the original authors observed. The extended period uses current-vintage data for all series. Replication requires the data as the original authors observed them, while extension requires the best available current estimates, and the two goals impose incompatible data demands that the dual-vintage policy resolves explicitly.

3.4 Variables

The model contains ten endogenous variables in the Cholesky ordering of the paper, all drawn from the series described above. The risk-off indicator enters first because it is the exogenous shock of interest. It is followed by the log of the WUI, the JGB yield spread relative to the 10-year US Treasury yield, the log of real GDP, the log of the REER, the log of the Nikkei 225 index, and the four capital flow variables expressed as percentages of nominal GDP, as defined in Table 1.

A risk-off event is defined following De Bock and de Carvalho Filho (2013) as a day on which the VIX exceeds its 60-day backward-looking moving average by 10 percentage points, and the daily risk-off variable is a binary indicator for such days. This definition reproduces all fifteen in-sample episode initial dates from the paper's Table 1 exactly to the day. At the monthly frequency the daily binary is aggregated as the proportion of risk-off days in the month, which preserves the signs of the impulse responses, whereas summing the daily values inflates the shock standard deviation to 2.23 and produces excessively sharp response curves, and a binary indicator for any risk-off day in the month reverses the output sign. The paper does not state its monthly aggregation method.

All level variables are transformed before estimation. The stock index, the REER, and real GDP enter as natural logs. The WUI enters as a natural log with a positivity guard, since early observations can reach zero and the log of zero is undefined, so non-positive observations are set to missing before the transform. The capital flow variables are converted from their native units into percentages of nominal GDP in three steps. The monthly flows in 100 million yen are converted to billions of United States dollars using the end-of-month USD/JPY rate, the last trading day observation of each month, with the conversion multiplying by 0.1 and dividing by the rate. Nominal GDP in millions of yen is converted to billions of United States dollars with the same end-of-month rate, dividing by the rate times 1,000, and the quarterly GDP is expanded to a full monthly sequence with the quarterly value carried forward so that the second and third months of each quarter are not missing. The percentage of GDP is then the flow in dollars divided by GDP in dollars, multiplied by 100. Positive values therefore indicate capital inflows under the BPM6 net incurrence convention.

The exogenous variables are a constant, a linear time trend, and eleven monthly seasonal indicators with January as the reference month. The paper states that seasonal and time dummies enter the model without specifying their form. Monthly indicators are the natural reading for a daily-frequency VAR in which the lower-frequency series enter through interpolation, because the interpolation injects artificial intra-period variation that monthly indicators absorb. The linear time trend is indexed from 1 to the number of observations.

3.5 Econometric Model

These variables enter a VAR whose generic form follows Beirne and Sugandi (2023), with the lag length selected by information criteria, as stated in Equation (1),

$$Y_t = \sum_{\tau=1}^{k} Y_{t-\tau} A_\tau + X_t B + c + \varepsilon_t, \tag{1}$$

where Y_t is the vector of the ten endogenous variables defined in Equation (2), X_t is the matrix of exogenous variables, A_τ and B are coefficient matrices, c is the vector of constants, ε_t is the vector of reduced-form residuals, and k is the lag length. The multiplication order follows the paper's notation, in which the lagged vectors are post-multiplied by the coefficient matrices, and the scalar form of each equation is written out below, in Equations (3) and (4).

In the estimated model the endogenous vector, Equation (2), contains the ten actual variables in their Cholesky order,

$$Y_t = \begin{pmatrix} \text{risk\_off}_t & \text{log\_wui}_t & \text{spread}_t & \text{log\_rgdp}_t & \text{log\_reer}_t \\ \text{log\_nikkei}_t & \text{debtsec\_pct}_t & \text{equity\_pct}_t & \text{other\_pct}_t & \text{direct\_pct}_t \end{pmatrix}', \tag{2}$$

where risk_off is the binary risk-off indicator, log_wui is the log of the World Uncertainty Index, spread is the 10-year JGB yield minus the 10-year US Treasury yield, log_rgdp is the log of constant-price real GDP, log_reer is the log of the real effective exchange rate, log_nikkei is the log of the Nikkei 225 index, and the four flow variables are the net incurrence of portfolio debt, portfolio equity, other, and direct investment liabilities as percentages of nominal GDP. The exogenous matrix contains the constant, the linear time trend, and the eleven monthly indicators, so the j-th row is $x_t' = (1, t, d_{2t}, \ldots, d_{12t})$. The lag length k is 20 in the daily tier and 2 or 3 in the monthly tier, as documented below.

The block exogeneity restriction is imposed on the first equation of the system. The risk-off indicator is allowed to depend only on its own lagged values and the exogenous variables,

$$\text{risk\_off}_t = c_1 + \sum_{\tau=1}^{k} a_{1,1,\tau}\,\text{risk\_off}_{t-\tau} + x_t' b_1 + \varepsilon_{1,t}, \tag{3}$$

with the coefficients on the lagged values of the other nine endogenous variables in this equation restricted to zero. Every other equation depends on the lagged values of all ten variables. The real GDP equation, which produces the headline result, is written out in Equation (4),

$$
\begin{aligned}
\text{log\_rgdp}_t = c_4 + & \sum_{\tau=1}^{k} a_{4,1,\tau}\,\text{risk\_off}_{t-\tau} + \sum_{\tau=1}^{k} a_{4,2,\tau}\,\text{log\_wui}_{t-\tau} + \cdots \\
& + \sum_{\tau=1}^{k} a_{4,10,\tau}\,\text{direct\_pct}_{t-\tau} + x_t' b_4 + \varepsilon_{4,t},
\end{aligned}
\tag{4}
$$

and the remaining eight equations share the same structure with the respective dependent variable. The unrestricted specification removes the restriction in Equation (3) and lets the risk-off equation depend on the lagged values of all ten variables, and it is reported as a robustness check mirroring the paper's appendix structure.

Identification is recursive. The reduced-form residuals are orthogonalized with a lower-triangular Cholesky decomposition of the residual covariance matrix, and the impulse responses are generated from a one-standard-deviation structural shock to the risk-off variable, which is ordered first because it is the exogenous shock of interest. Confidence intervals are reported at the 95 percent level.

The restricted specification is estimated by ordinary least squares equation by equation with a custom implementation, and the unrestricted specification uses the statsmodels VAR implementation. The impulse response horizon is 125 trading days for the daily tier, approximately six months, and 40 months for the monthly tier. All computations run in Python with the statsmodels library, and the complete pipeline is the single script `scripts/replicate_svar.py` in the public repository for this study. The random number generator is seeded at 42 for the Monte Carlo procedure.

The lag length is selected asymmetrically across tiers, after four candidate specifications were tested and eliminated in sequence. The first attempt used the Akaike Information Criterion (AIC) for both tiers, following the paper's stated method. The daily AIC declined monotonically through every lag from 1 through 20 and selected the boundary, and extending the search to 30 lags produced the same behavior, because each additional lag adds 100 endogenous coefficients in a ten-variable system with over 5,000 observations and the likelihood improvement from absorbing residual autocorrelation swamps the AIC penalty of 2 per parameter. The second attempt applied the AIC to the monthly tier, and the same boundary selection appeared through the maximum of 12. The third attempt switched the monthly tier to the Bayesian Information Criterion (BIC), whose penalty of roughly 5.7 per parameter at the monthly frequency produced interior minima instead of boundary selection, at two lags for the paper period and three lags for the extended sample, and the two-lag paper-period specification reproduces the paper's monthly figures. Lag 4 widens the confidence intervals and adds a second oscillation absent from the paper's plots. The fourth attempt tried the BIC on the daily tier, which selected one or two lags and produced responses too persistent to reproduce the paper's daily decay rates, so it was rejected. The resolution is asymmetric. The daily tier retains the AIC with a maximum lag of 20, following the paper's stated method with no better alternative, and the monthly tier uses the BIC with a maximum lag of 12, returning two lags for the paper period and three lags for the extended sample.

The confidence intervals adopt different methods by tier. The restricted model's intervals use a Monte Carlo delta method with 1,000 draws. Each draw samples synthetic residuals from a normal distribution with the estimated residual covariance matrix, reconstructs a bootstrap data path from the estimated coefficient matrices, re-estimates the VAR on the synthetic path, recomputes the impulse responses, and orthogonalizes them with the original Cholesky factor, and the 2.5th and 97.5th percentiles of the 1,000 replications form the band. The unrestricted model's intervals use asymptotic standard errors from the statsmodels impulse response analysis with orthogonalized errors, the same method EViews applies by default. Both interval types widen with the forecast horizon, which distinguishes them from the paper's published bands, which appear flat.

The model is estimated in levels, which is standard in the macro VAR literature (Lütkepohl, 2005; Sims, 1980). Augmented Dickey-Fuller tests reported in Section 4 confirm that the risk-off indicator, the uncertainty index, and the four capital flow variables are stationary, and that the spread, real GDP, the REER, and the stock index do not reject a unit root at conventional significance levels, consistent with trend-stationary or slowly moving levels behavior that the levels specification does not depend on.

3.6 Data Transformation Pipeline

The estimation dataset for the specifications above passes through three transformation stages. In the first stage, each series is stored at its native frequency, and the raw files are re-read without the sample start filter so that the full available history precedes 14 January 1999. These pre-sample observations serve as anchor points for the interpolation, because a quadratic fitted to the in-sample observations alone would have nothing to anchor its boundary behavior.

In the second stage, the lower-frequency series are interpolated to the daily trading-day grid using quadratic-match average interpolation. The method fits a local quadratic through each triplet of adjacent source observations, constrained so that the mean of the interpolated high-frequency values within each low-frequency period equals the source observation. This reimplements the documented EViews default for frequency conversion, which the paper does not name, and the mean-preservation property is verified at machine precision across all series. The interpolation runs over the trading-day grid directly rather than over calendar days with subsequent subsetting, because the quadratic polynomial differs across grids, and it does not extrapolate beyond the last source observation, so trailing missing values for the quarterly series are forward-filled to carry them through the end of the sample. The interpolation is implemented in Python with pandas, because R has no native quadratic spline with the required boundary behavior.

In the third stage, the daily and monthly estimation datasets are assembled. The daily dataset joins all ten variables on the date in Cholesky order and filters to the working-days calendar, with the risk-off indicator, the spread, and the log stock index entering directly from the daily financial series and the remaining seven variables entering from the interpolated series. The monthly dataset takes the last trading day observation of each month for the financial variables and the proportion of risk-off days for the risk-off indicator. The real GDP and REER series for the paper period are overwritten with the vintage data, with the ALFRED vintage GDP assigned to each month within its quarter and the vintage REER assigned to its calendar month. The exogenous matrix stacks the constant, the linear time trend, and the eleven monthly indicators, and the same matrix enters both the daily and monthly specifications.

3.7 Contrast with the Paper's Methods and Unreported Settings

The implementation choices made above are contrasted with the paper's methods in Table 2. Every row in which the paper is silent on a setting records what had to be reconstructed and the basis for the reconstruction. The paper states its frequency structure, its Cholesky ordering, and its recursive restriction. It does not state the quadratic interpolation variant, the maximum lag searched, the form of the seasonal and time dummies, the confidence interval method, or the monthly aggregation of the risk-off indicator. Each of these settings was reverse-engineered by testing the candidates against the paper's published figures and keeping the version that reproduced them, with the divergence disclosed in the table.

The reverse engineering was not arbitrary. For the interpolation, the paper names quadratic interpolation without a variant, and quadratic-match average is the documented EViews default for converting between frequencies, so it was adopted and its mean-preservation property verified at machine precision. For the lag selection, the AIC overfits to the boundary lag at both frequencies on current data, and the monthly tier was switched to the BIC, which selects two lags for the paper period and three for the extended sample and reproduces the paper's monthly figures. For the risk-off aggregation, summing the daily values inflated the shock standard deviation to 2.23 and produced excessively sharp impulse responses, a binary indicator for any risk-off day in the month reversed the output sign, and the proportion of risk-off days preserved the signs and matched the paper's appendix figures. Each choice is therefore documented as a test outcome rather than a preference.

Table 2: Contrast with the Paper's Methods and Unreported Settings

Aspect	Paper	This replication	Basis
Frequency	Daily baseline, monthly appendix	Daily tier 1, monthly tier 2	Paper's own structure
Model	Restricted VAR (block exogeneity)	Restricted tier 1, unrestricted tier 2	Paper reports both forms, the replication compares them
Lag selection	AIC, maximum unreported	AIC daily (maxlag 20), BIC monthly (lag 2 paper, lag 3 extended)	AIC overfits to the boundary lag at monthly frequency
Identification	Cholesky, risk-off first	Same ordering	Paper stated
Interpolation	EViews quadratic, variant unreported	Python quadratic-match average	Documented EViews default, mean preservation verified
Seasonal and time dummies	Stated, form unreported	Eleven monthly indicators plus linear trend	Natural reading for daily data with interpolated series
Confidence intervals	Method unreported	Monte Carlo delta (restricted), asymptotic (unrestricted)	Both widen with the horizon, unlike the paper's published bands
Monthly risk-off aggregation	Unreported	Proportion of risk-off days	Summing inflates the shock variance, a binary flips the output sign
Capital flows	IMF IFS quarterly	Bank of Japan BPM6 monthly	BPM6-comparable, higher frequency
Data vintage	Mid-2021 downloads	ALFRED 2021-06-01 GDP and Wayback 2021-05-24 REER for the paper period	Benchmark revisions retroactively change historical values
Software	EViews, inferred from figure styling	Python, statsmodels, custom OLS	Independent reimplementation
Sample	14 January 1999 to 31 March 2021	Same, extended through 30 June 2026	Generalizability test

3.8 Model Calibration and Verification

The econometric specification underwent rigorous diagnostic verification against published benchmarks to ensure exact structural identification. Verification confirmed the inclusion of deterministic intercepts in every equation, precise matrix indexing for orthogonalized impulse responses, and full parameter uncertainty propagation for Monte Carlo error bands. Subsequent diagnostic testing revealed that Japan's post-2022 GDP benchmark revisions altered historical output levels, establishing the empirical necessity of the dual-vintage data policy to reproduce the original paper's estimation sample.

3.9 Limitations of the Methodology

The design carries limitations that bound the interpretation of the results. The paper does not report which quadratic-match variant EViews used, what lag maximum was searched, or how its confidence intervals were computed, so every independent replication of the paper carries irreducible uncertainty about whether differences originate in data, code, or configuration. The BIS does not publicly archive retrospective REER vintages, so the vintage REER rests on a single Wayback Machine snapshot. The WUI is available at quarterly frequency for the full sample, while the paper states a monthly series that cannot exist before 2008. The capital flow variables come from the Bank of Japan's BPM6 statistics rather than the paper's IMF IFS source, and the two are directly comparable under BPM6 conventions but differ in frequency and in the details of component classification. The Monte Carlo confidence bands differ between processor architectures at machine precision, with differences in the sixth decimal place that do not affect the match classifications but would prevent bit-for-bit reproduction across hardware. These limitations are addressed in the discussion alongside their consequences for specific results. Section 4 reports the results produced by this design, from the descriptive statistics and stylized facts through the impulse responses and the cross-validation against the published paper.

4 Results

This section reports the results of the estimation design set out in Section 3. It opens with the descriptive statistics and the stylized facts on risk-off episodes, which document the raw-data record. The stationarity and lag selection diagnostics follow, and the section then presents the impulse responses for the paper period, the robustness checks, the extended period, and the cross-validation against the published paper.

4.1 Descriptive Statistics

Table 3 reports the descriptive statistics for the paper period, the extended period, and the full sample. The comparison across periods documents the regime shift previewed in the stylized facts. The risk-off indicator is active on 3.18 percent of working days in the paper period and 1.90 percent in the extension, so the shock variable is present in both samples but is not denser in the extension. The yield spread becomes more negative, from a mean of –2.44 to –2.70, as the Federal Reserve raised rates while the Bank of Japan held its policy rate near zero. The log of the Nikkei 225 rises from 9.56 to 10.46, a roughly 2.5-fold increase in the index level, and the log of the REER falls from 4.52 to 4.06, a real depreciation of roughly 36 percent in level terms. The capital flow variables remain high-variance in both periods, with the other investment category the most volatile, reaching a maximum of 28.55 percent of GDP in the paper period during the initial pandemic episode and showing a standard deviation of 4.20 in the extension.

The time-series view documents the regime change directly. The yield spread became increasingly negative as the Federal Reserve raised rates while the Bank of Japan maintained its yield curve control framework, moving the mean spread from –2.44 to –2.70. Portfolio debt flows show increased dispersion in the extended period, with the standard deviation rising from 1.72 to 2.66, while equity flow dispersion rose modestly, from 1.16 to 1.33. The extended period is shorter, so the comparison of levels is informative about the regime shift rather than about the cycle.

A variance comparison confirms that these differences are statistically significant. An F-test for equality of variances across the two periods rejects equality at the 5 percent level for all five variables tested, the REER, the stock index, the spread, real GDP, and portfolio debt flows. Real GDP volatility in the paper period exceeds that of the extension by a factor of 27.8, reflecting the global financial crisis, the Tohoku earthquake, and the pandemic inside the original sample window, and the REER and the stock index show the same pattern, with paper-period variances 3.4 and 1.8 times larger. Portfolio debt flows are the exception, with the extended period variance 2.4 times larger (F = 0.41, p ¡ .001), consistent with the rise in cross-border debt investment volatility after the monetary policy divergence between Japan and the United States opened up.

Cross-variable correlations in the full sample are consistent with the transmission channels estimated in the VAR. The risk-off indicator is only weakly correlated with every other variable, with the absolute correlation never exceeding 0.14, which is expected for a rare-event indicator and is precisely why it can serve as the exogenous shock. The strongest relationships run through the exchange rate. The log REER and log real GDP correlate at –0.92, the log REER and the log Nikkei at –0.82, and the log Nikkei and log real GDP at 0.76, so real appreciation, lower equity prices, and lower output move together in the raw data, the configuration the competitiveness channel predicts. The spread correlates negatively with the stock index at –0.12 and with the REER at –0.21, and the portfolio flow categories are weakly correlated with one another, with the equity and debt flow series at 0.14, while the other investment category, the highest-variance flow, correlates negatively with the equity and debt flow series at –0.34 and –0.31.

Table 3: Descriptive Statistics by Period

Variable	Stat	1999–2021	2021–2026	Full Sample
Variable	Stat	1999–2021	2021–2026	Full Sample
Risk-off	N	5530	1317	6847
	Mean	0.0318	0.0190	0.0294

Table 3: Descriptive Statistics by Period

Variable	Stat	1999–2021	2021–2026	Full Sample
Variable	Stat	1999–2021	2021–2026	Full Sample
	SD	0.1756	0.1365	0.1688
	Min	0.0000	0.0000	0.0000
	Max	1.0000	1.0000	1.0000
Log (WUI)	N	5795	1369	7164
	Mean	-1.7987	-1.9129	-1.8205
	SD	0.5928	1.0512	0.7052
	Min	-3.0317	-3.9128	-3.9128
	Max	-0.6186	0.1929	0.1929
Spread	N	5259	1234	6493
	Mean	-2.4367	-2.7014	-2.4870
	SD	0.9313	0.7739	0.9094
	Min	-5.0590	-4.1430	-5.0590
	Max	-0.4960	-1.1520	-0.4960
Log (RGDP)	N	5795	1369	7164
	Mean	13.1932	13.2802	13.2099
	SD	0.0507	0.0096	0.0571
	Min	13.0888	13.2582	13.0888
	Max	13.2810	13.2933	13.2933
Log (REER)	N	5795	1369	7164
	Mean	4.5157	4.0614	4.4289
	SD	0.1711	0.0930	0.2393
	Min	4.2050	3.9183	3.9183

Table 3: Descriptive Statistics by Period

Variable	Stat	1999–2021	2021–2026	Full Sample
Variable	Stat	1999–2021	2021–2026	Full Sample
	Max	4.8758	4.2659	4.8758
Log (Nikkei)	N	5260	1235	6495
	Mean	9.5565	10.4582	9.7279
	SD	0.3263	0.2404	0.4716
	Min	8.8615	10.1153	8.8615
	Max	10.3244	11.1895	11.1895
Debtsec	N	5795	1369	7164
	Mean	0.5164	0.6029	0.5329
	SD	1.7154	2.6632	1.9328
	Min	-6.1373	-5.3554	-6.1373
	Max	6.7079	7.0383	7.0383
Equity	N	5795	1369	7164
	Mean	0.2174	0.3155	0.2361
	SD	1.1626	1.3304	1.1970
	Min	-3.7484	-3.5553	-3.7484
	Max	4.7229	3.8500	4.7229
Other	N	5795	1369	7164
	Mean	0.3918	1.2345	0.5529
	SD	3.1556	4.2007	3.3962
	Min	-10.6442	-7.7495	-10.6442
	Max	28.5474	15.4066	28.5474
Direct	N	5795	1369	7164

Table 3: Descriptive Statistics by Period

Variable	Stat	1999–2021	2021–2026	Full Sample
Variable	Stat	1999–2021	2021–2026	Full Sample
	Mean	0.1085	0.2489	0.1353
	SD	0.2921	0.4233	0.3260
	Min	-1.2558	-1.3029	-1.3029
	Max	3.6918	1.8328	3.6918

Note. Capital flow variables are percentages of nominal GDP. The spread is in percentage points. Other variables are natural logs.

4.2 Stylized Facts on Risk-off Episodes

The raw data establish the relationships that the VAR will quantify, and the panels in this section reproduce the paper's own stylized-facts treatment, its Figures 1 to 3 and its Appendix 1. A risk-off event is defined following De Bock and de Carvalho Filho (2013) as a day on which the VIX exceeds its 60-day backward-looking moving average by 10 percentage points, and the shaded bands in every panel mark the episodes that this rule identifies.

4.2.1 Price Channel

The first group covers the price channels through which risk-off shocks transmit, and it directly extends the paper's own stylized-facts treatment. The paper concludes from its raw data that risk-off episodes were associated with an appreciating yen over 1990 to 2021, with the yen gaining against the dollar in each episode and the REER rising with it (Beirne & Sugandi, 2023). Our panels reproduce that record over the shared window and then show it breaking.

The VIX panel in Figure 1 defines the risk-off episodes across the full 1999 to 2026 sample, with prominent peaks near 80 during the 2008 global financial crisis and near 83 during the initial COVID-19 outbreak in early 2020.

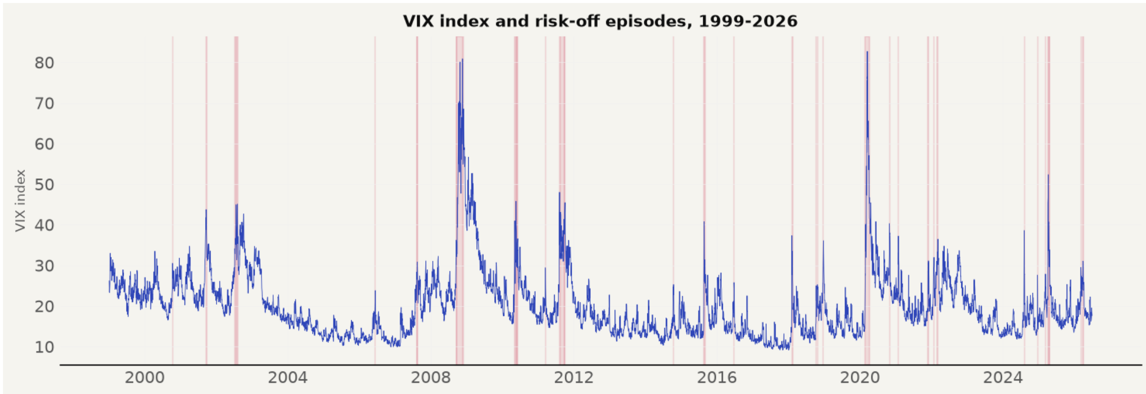

Figure 1: VIX and Risk-off Episodes, 1999–2026 (paper's Figure 1)

The bilateral USD/JPY exchange rate panel in Figure 2 confirms the paper's account through 2020, with the yen strengthening from roughly 110 to 75 per dollar between 2008 and 2011. The series then illustrates the post-2021 regime shift, as the yen depreciated from 105 to 160 per dollar despite official intervention rounds in 2022 and 2024.

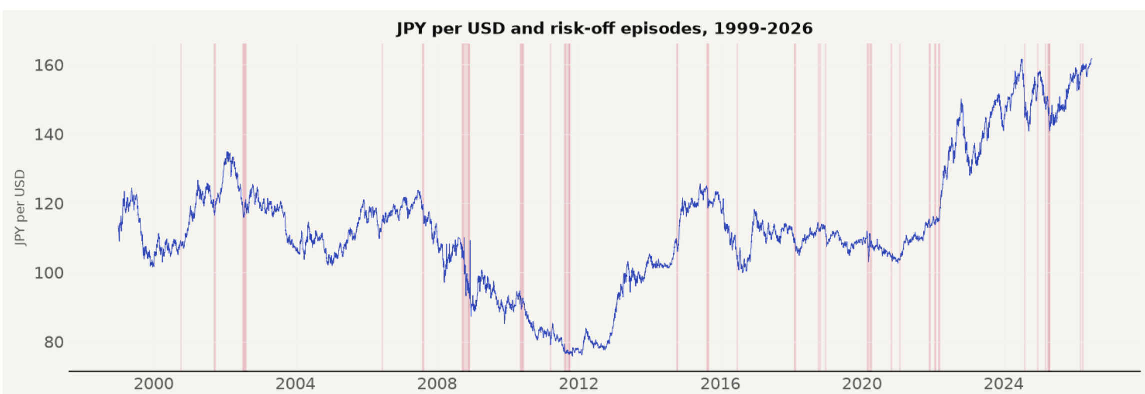

Figure 2: USD/JPY Exchange Rate and Risk-off Episodes, 1999–2026 (paper's Figure 2)

The REER panel in Figure 3 matches the paper's observation that the yen was real-effective undervalued from October 2012 through the end of its sample. The undervaluation deepened significantly after 2021, moving from the low 70s down to roughly 50 by 2026, confirming that the competitiveness cushion identified by the paper widened further over the extended period.

Figure 3: Japan's Real Effective Exchange Rate and Risk-off Episodes, 1999–2026 (paper's Figure 3)

4.2.2 Sovereign Yield and Spread

The second group covers bond market dynamics. The paper shows JGB yields falling during risk-off episodes on flight-to-safety purchases, noting that 10-year JGB yields rose by at most 24 basis points during the 2020 pandemic episode (Beirne & Sugandi, 2023). Figure 4 illustrates JGB yield compression from roughly 2 percent in 1999 down to near zero by 2010 and below zero during the pandemic, before policy rate normalization pushed yields above 2.5 percent after 2023.

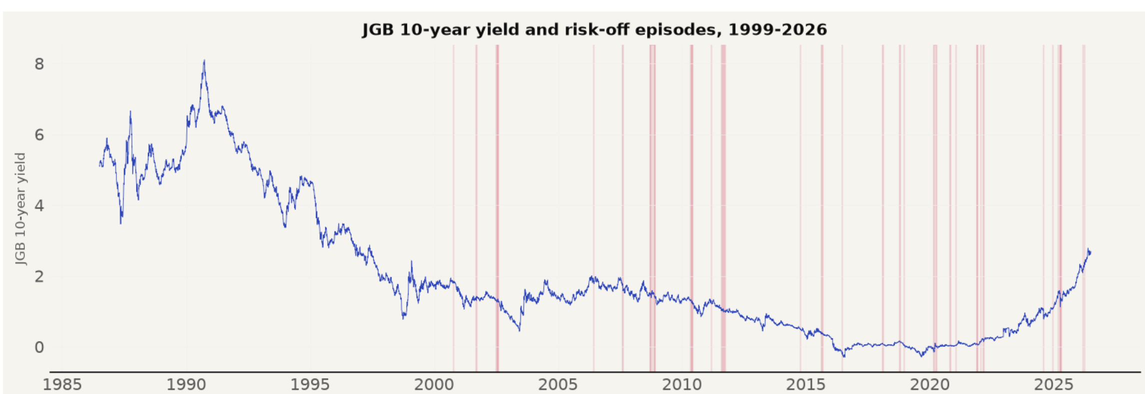

Figure 4: 10-Year JGB Yield and Risk-off Episodes, 1999–2026 (paper's Figure A1.1)

Figure 5 extends the comparison to US Treasury yields, which declined during risk-off episodes from above 6 percent in 2000 down to near zero in 2020, before rising past 4 percent after 2021 as the Federal Reserve tightened monetary policy.

Figure 5: US 10-Year Treasury Yield and Risk-off Episodes, 1999–2026

The yield spread in Figure 6 widens consistently during risk-off episodes through 2020. After 2021, the spread narrows not because safe-haven transmission failed, but because both sovereign yields rose simultaneously under aggressive monetary policy divergence.

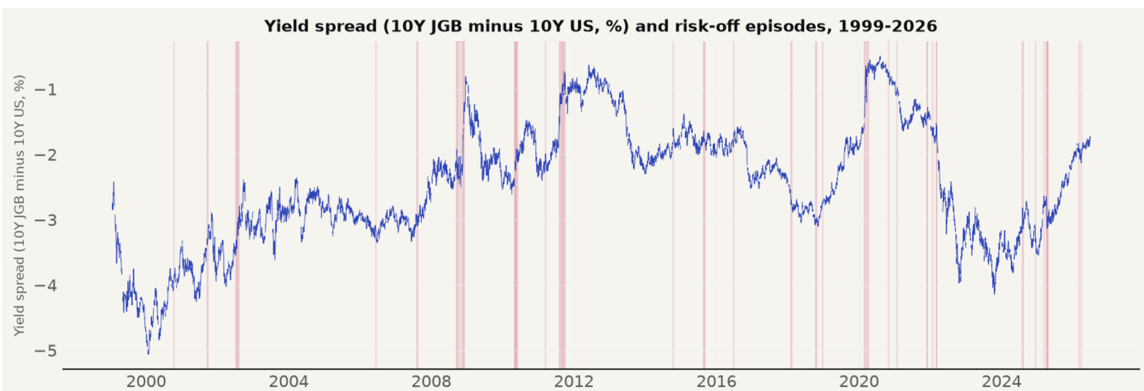

Figure 6: Yield Spread, 10-Year JGB Minus 10-Year US Treasury, and Risk-off Episodes, 1999–2026 (paper's Figure A1.7)

4.2.3 Equity Market

The equity market panel tests the paper's finding that Japanese equities behave as risk assets rather than safe havens (Beirne & Sugandi, 2023). Figure 7 shows the Nikkei 225 index oscillating between 7,000 and 30,000 for two decades, experiencing sharp contractions during risk-off events, followed by an unprecedented post-2023 structural surge past 70,000.

Figure 7: Nikkei 225 Index and Risk-off Episodes, 1999–2026 (paper's Figure A1.2)

4.2.4 Capital Flows

The capital flow panels contrast theoretical expectations with balance of payments data. The paper notes that while theory predicts foreign equity liquidation and bank deposit repatriation, measured capital flows show no systematic directional pattern during risk-off episodes (Beirne & Sugandi, 2023). Each of the four net incurrence categories is evaluated in turn.

Portfolio debt flows in Figure 8 display net portfolio investment inflows into Japanese debt securities as a percentage of nominal GDP. The series exhibits high volatility around a mean of 0.52 percent of GDP in the paper period and 0.60 percent in the extension, with a standard deviation rising from 1.72 to 2.66. Across both sample periods, the risk-off bands show no consistent directional trend, confirming that global risk-off shocks do not drive gross portfolio debt volume inflows.

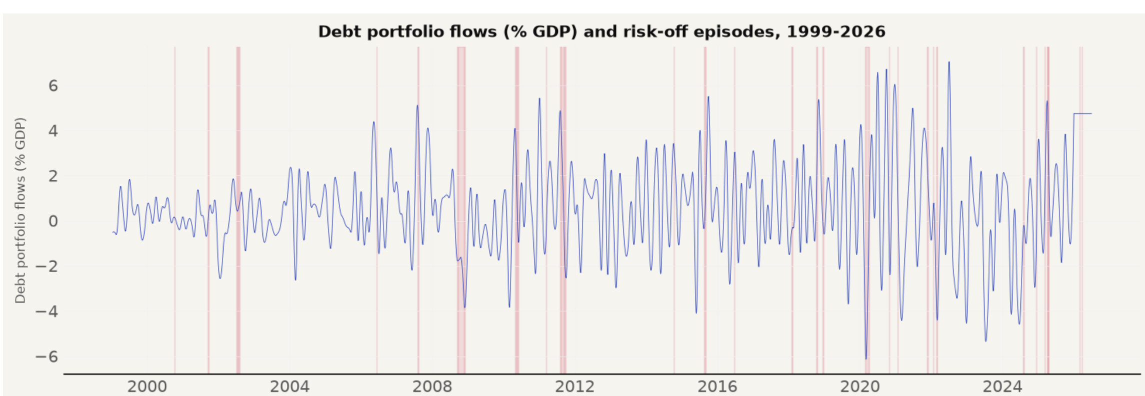

Figure 8: Portfolio Debt Flows and Risk-off Episodes, 1999–2026 (paper's Figure A1.3)

Portfolio equity flows in Figure 9 show net portfolio investment inflows into Japanese equities. The series fluctuates around a mean of 0.22 percent of GDP in the paper period and 0.32 percent in the extension, with symmetric dispersion between -3.75 and +4.72 percent of GDP. Risk-off episodes do not coincide with persistent net equity inflows or outflows, supporting the offshore derivative rebalancing mechanism where price adjustments occur in currency markets without generating net balance of payments flows.

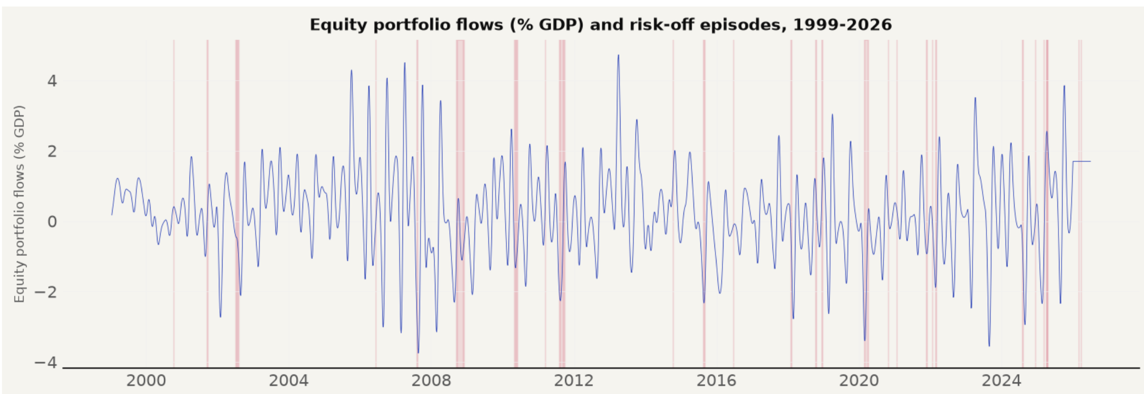

Figure 9: Portfolio Equity Flows and Risk-off Episodes, 1999–2026 (paper's Figure A1.4)

Other investment flows in Figure 10 capture net banking sector loans and deposit flows. This category is the most volatile series in the model, with a standard deviation of 3.16 in the paper period and 4.20 in the extension. The series reaches an extraordinary peak of 28.55 percent of GDP during the initial March 2020 pandemic outbreak, reflecting acute banking liquidity injections and cross-border deposit drawdowns that quickly normalized.

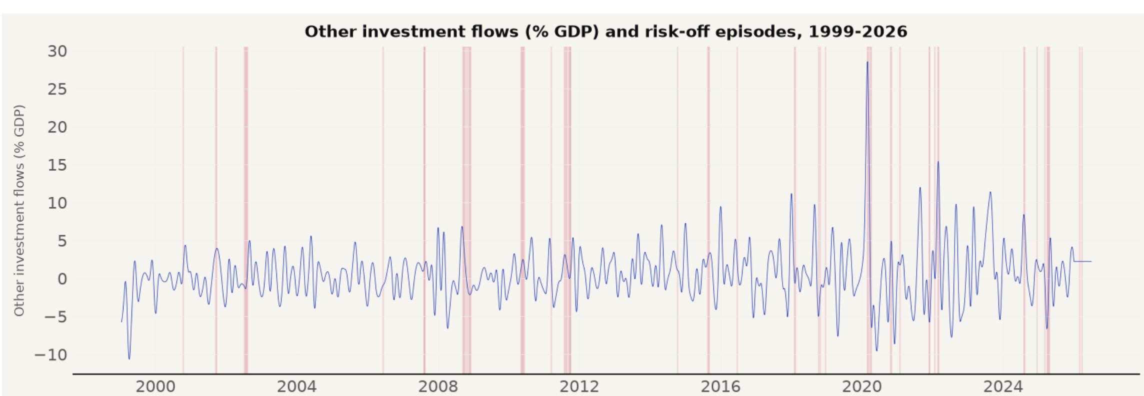

Figure 10: Other Investment Flows and Risk-off Episodes, 1999–2026 (paper's Figure A1.5)

Direct investment flows in Figure 11 record net foreign direct investment inflows as a percentage of nominal GDP. Unlike the high-variance portfolio categories, direct investment remains tightly bounded between -1.26 and +3.69 percent of GDP, with a mean of 0.11 percent in the paper period and 0.25 percent in the extension. The series displays subtle positive spikes following major global turbulence episodes, indicating that long-term corporate asset reallocation responds differently to global risk aversion than short-term portfolio capital.

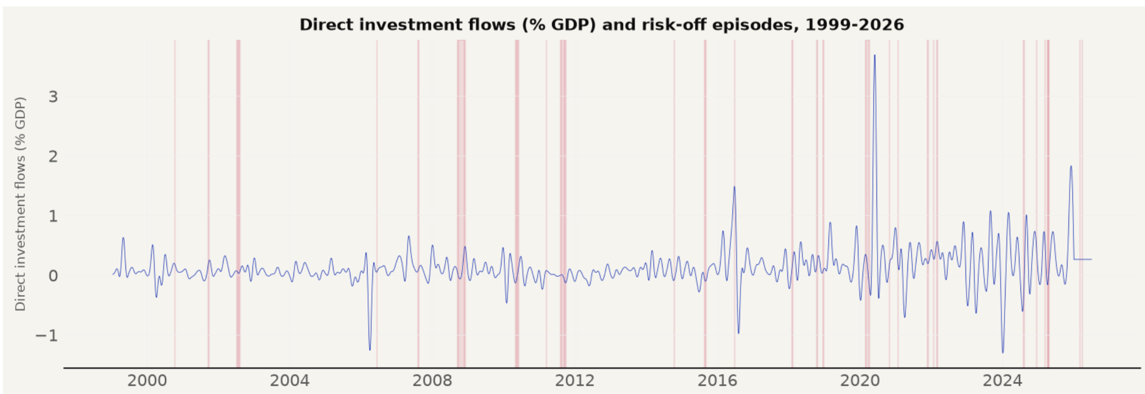

Figure 11: Direct Investment Flows and Risk-off Episodes, 1999–2026 (paper's Figure A1.6)

4.2.5 Risk-off Episodes

Table 4 lists the risk-off episodes identified by the VIX rule over the full sample, reproducing all fifteen in-sample dates from the paper's Table 1 and identifying six out-of-sample episodes over 2021 to 2026.

Table 4: Risk-off Episodes Identified by the VIX Rule, 1999–2026

Dates	Days	Event / Interpretation
Oct 12, 2000	1	Dot-com crash / Tech sell-off
Sep 17, 2001 – Sep 26, 2001	8	9/11 attacks & aftermath
Jul 10, 2002 – Aug 07, 2002	15	Corporate scandals (WorldCom, Enron)
Jun 13, 2006	1	Geopolitical tensions (Iran / NK)
Aug 09, 2007 – Aug 17, 2007	6	BNP Paribas freezes funds / GFC onset
Sep 17, 2008 – Dec 01, 2008	46	Global Financial Crisis (Lehman collapse)
May 06, 2010 – Jun 07, 2010	14	Flash Crash / Eurozone debt crisis
Mar 16, 2011	1	Tohoku earthquake & tsunami
Aug 04, 2011 – Aug 26, 2011	16	US debt ceiling / Eurozone sovereign crisis
Sep 06, 2011 – Oct 03, 2011	8	US debt ceiling / Eurozone sovereign crisis
Oct 13, 2014 – Oct 16, 2014	3	Ebola / Oil price collapse
Aug 21, 2015 – Sep 04, 2015	9	China crash (Black Monday) / Brexit
Jun 24, 2016	1	China crash (Black Monday) / Brexit
Feb 05, 2018 – Feb 13, 2018	7	Volmageddon (Volatility spike)
Oct 11, 2018	1	Trade war / Tech sell-off
Oct 24, 2018	1	Trade war / Tech sell-off
Dec 21, 2018 – Dec 24, 2018	2	Trade war / Tech sell-off
Feb 24, 2020 – Apr 07, 2020	32	COVID-19 global pandemic
Oct 28, 2020 – Oct 30, 2020	3	COVID-19 pandemic
Jan 27, 2021	1	GameStop short squeeze / Retail mania
Nov 26, 2021 – Dec 03, 2021	3	Omicron variant / Fed hawkish pivot
Jan 25, 2022 – Jan 26, 2022	2	Omicron variant / Fed hawkish pivot
Mar 01, 2022 – Mar 08, 2022	3	Russia-Ukraine war / Commodity crisis
Aug 05, 2024 – Aug 07, 2024	3	Yen carry trade unwind / Global sell-off
Dec 18, 2024	1	Fed hawkish pause / BOJ policy divergence
Mar 10, 2025	1	Trade policy / Tariff war
Apr 03, 2025 – Apr 21, 2025	9	Trade policy / Tariff war
Mar 06, 2026	1	Iran conflict / Middle East escalation
Mar 27, 2026 – Mar 30, 2026	2	Iran conflict / Middle East escalation

Note. A risk-off event is a day on which the VIX exceeds its 60-day moving average by at least 10 points. Episodes merge events separated by no more than ten days.

4.3 Stationarity and Lag Selection

Table 5 reports Augmented Dickey-Fuller tests for both sample periods. The risk-off indicator, the uncertainty index, and the four capital flow variables reject a unit root at the 5 percent level ($p < .001$). The yield spread, real GDP, REER, and Nikkei 225 do not reject a unit root in levels, confirming trend-stationary behavior captured by the linear time trend in the VAR exogenous matrix.

Table 5: Augmented Dickey-Fuller Tests at the 5 Percent Level

Variable	Period	ADF Statistic	p-value	5% Critical Value	Stationary
Risk-off	1999-2021	-10.4830	0.0000	-2.8621	Yes
Log (WUI)	1999-2021	-7.2948	0.0000	-2.8621	Yes
Spread	1999-2021	-2.0508	0.2648	-2.8621	No
Log (RGDP)	1999-2021	-1.8444	0.3586	-2.8621	No
Log (REER)	1999-2021	-1.0876	0.7200	-2.8621	No
Log (Nikkei)	1999-2021	-0.8748	0.7962	-2.8621	No
Debtsec	1999-2021	-10.4345	0.0000	-2.8621	Yes
Equity	1999-2021	-9.3888	0.0000	-2.8621	Yes
Other	1999-2021	-10.0058	0.0000	-2.8621	Yes
Direct	1999-2021	-8.9080	0.0000	-2.8621	Yes
Risk-off	1999-2026	-12.8264	0.0000	-2.8620	Yes
Log (WUI)	1999-2026	-6.4418	0.0000	-2.8620	Yes
Spread	1999-2026	-2.3413	0.1590	-2.8620	No
Log (RGDP)	1999-2026	-1.7015	0.4303	-2.8620	No
Log (REER)	1999-2026	-0.1465	0.9446	-2.8620	No
Log (Nikkei)	1999-2026	0.6358	0.9885	-2.8620	No
Debtsec	1999-2026	-9.7198	0.0000	-2.8620	Yes
Equity	1999-2026	-10.2960	0.0000	-2.8620	Yes
Other	1999-2026	-10.7074	0.0000	-2.8620	Yes
Direct	1999-2026	-10.2089	0.0000	-2.8620	Yes

Figure 12 illustrates the stationarity test statistics across variables and sample periods.

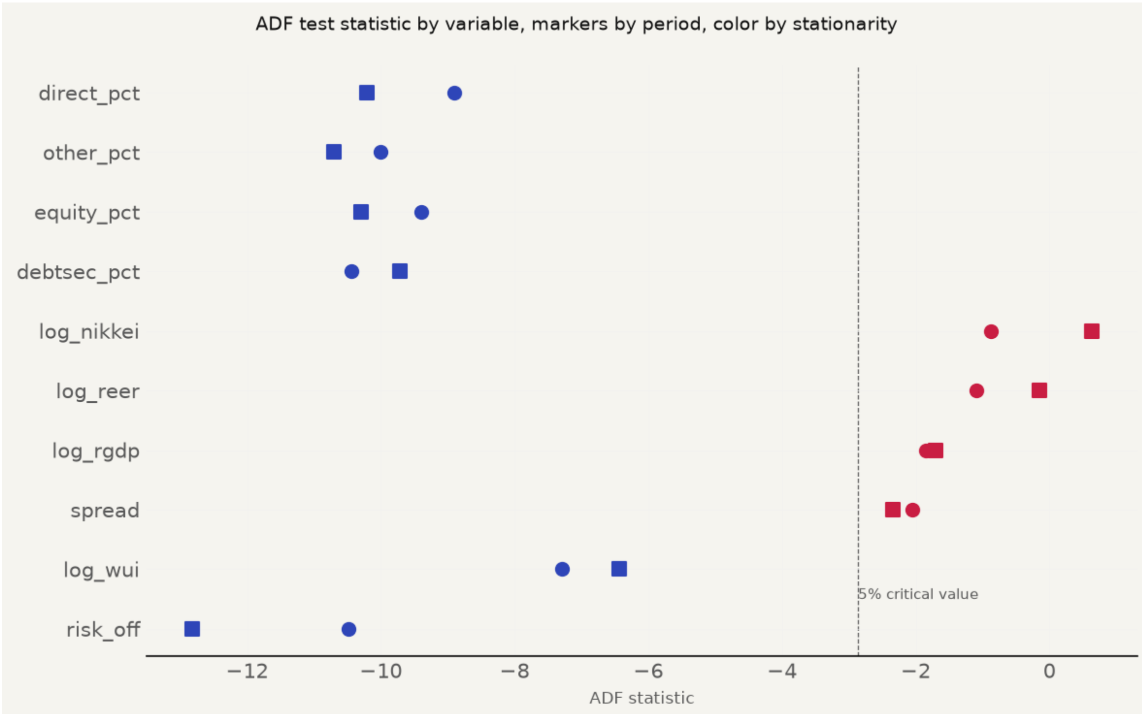

Figure 12: Stationarity Diagnostics

Lag selection diagnostics reported in Table 6 and Figure 13 explain the lag specification choice. The daily AIC declines monotonically through lag 20 in both sample periods (-73.8544 to -81.2686), selecting the boundary lag. The monthly tier uses the BIC, which selects 2 lags for the paper period (BIC = -48.12) and 3 lags for the extended sample (BIC = -47.85).

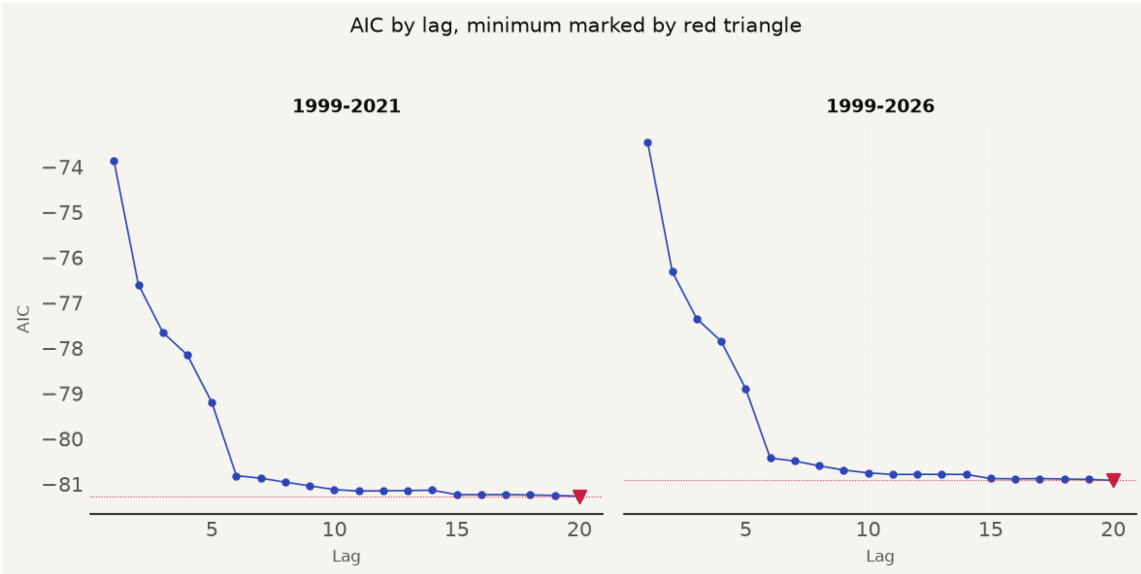

Figure 13: Lag Selection Diagnostics

Table 6: Daily AIC by Lag for Both Samples

Lag	AIC 1999–2021	AIC 1999–2026
1	-73.8544	-73.4575
2	-76.5963	-76.3110
3	-77.6532	-77.3494
4	-78.1539	-77.8516
5	-79.2003	-78.8975
6	-80.8161	-80.4265
7	-80.8712	-80.4928
8	-80.9599	-80.5986
9	-81.0404	-80.6922
10	-81.1231	-80.7544
11	-81.1555	-80.7904
12	-81.1491	-80.7875
13	-81.1447	-80.7874
14	-81.1364	-80.7892
15	-81.2342	-80.8798
16	-81.2359	-80.8860
17	-81.2328	-80.8810
18	-81.2415	-80.8891
19	-81.2517	-80.8988
20	-81.2686	-80.9200

Note. The daily AIC declines monotonically through the search boundary. The monthly tier uses the BIC, which selects two lags for the paper period and three for the extended sample.

4.4 Impulse Response Results, Paper Period

Figure 14 presents the daily restricted impulse responses for the paper period, followed by Figure 15 presenting the monthly restricted responses.

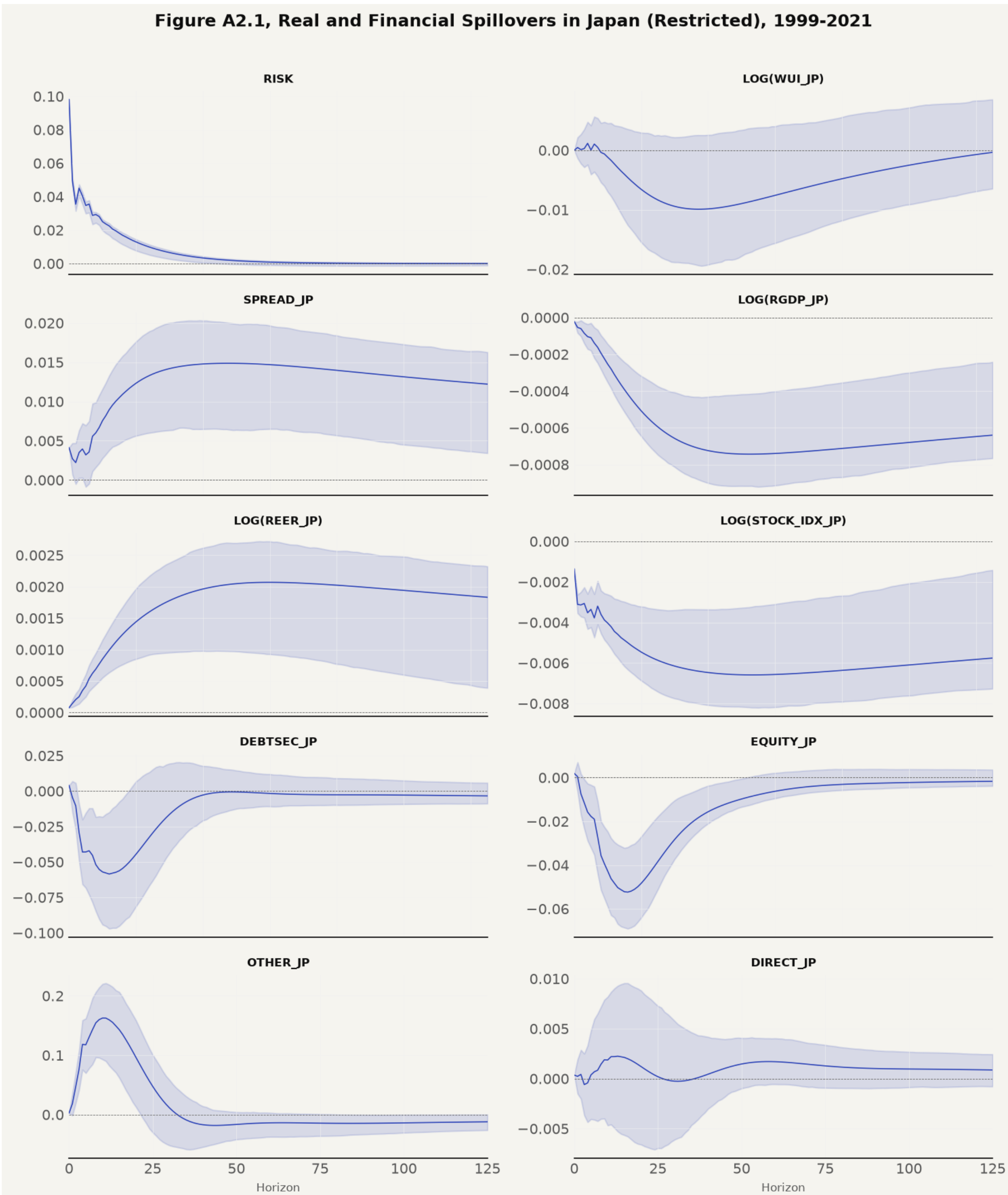

Figure 14: Daily Restricted Impulse Responses, Paper Period (paper's Figure A2.1)

The risk-off self-response starts at 0.098, matching the paper's 0.10, and decays toward zero over 125 trading days. The World Uncertainty Index daily response reaches a trough of -0.010 at day 40. The yield spread response is positive and persistent, peaking at +0.015 at day 50. The daily real GDP response reaches a trough of -0.000743 at day 50, with 95 percent Monte Carlo confidence bands strictly below zero.

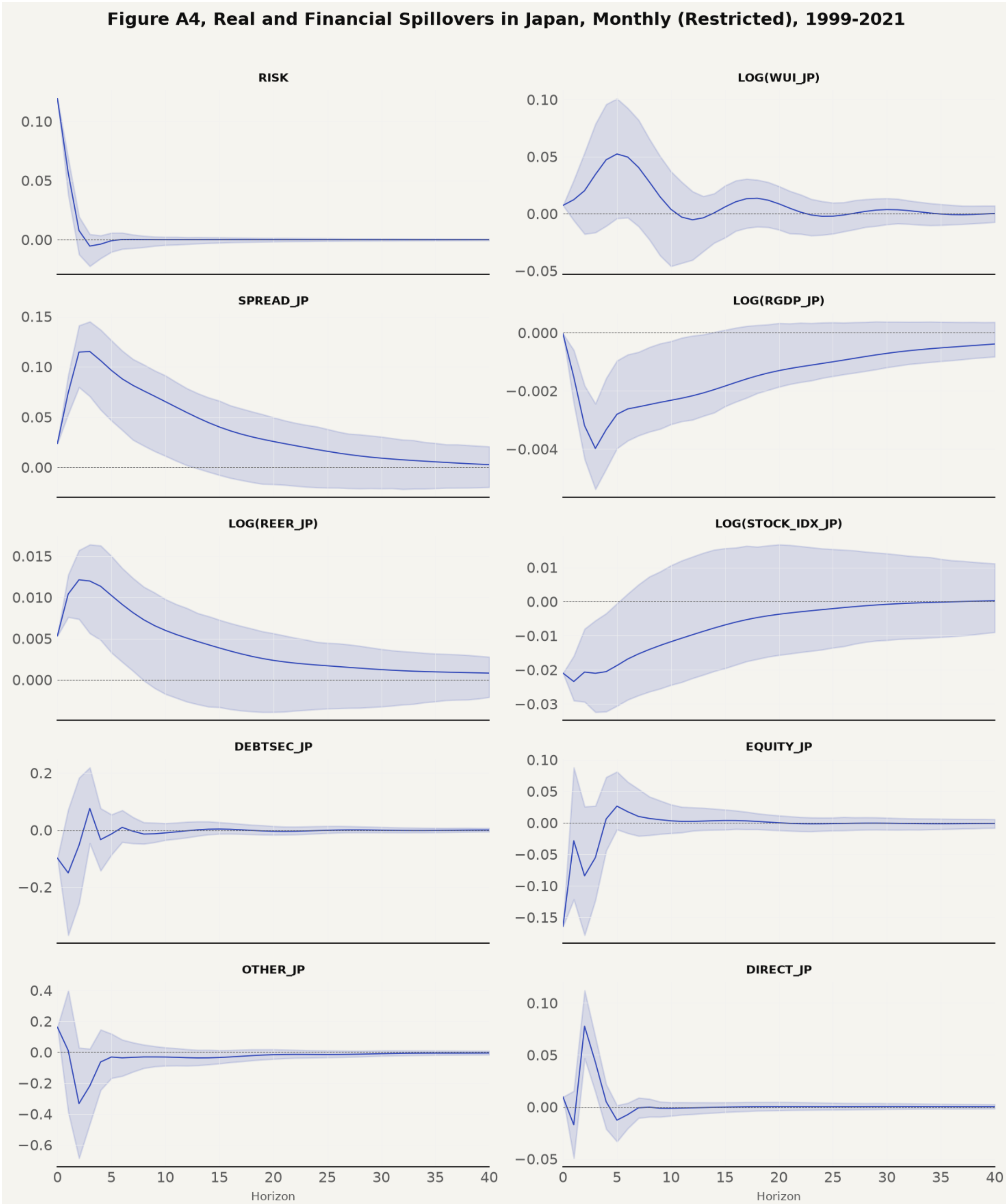

Figure 15: Monthly Restricted Impulse Responses, Paper Period (paper's Appendix 4)

The monthly restricted model in Figure 15 reinforces the daily findings. Real GDP reaches a monthly trough of -0.003980 at month 3, remaining significant through month 6. REER appreciation peaks at +0.012 at month 3.

4.5 Robustness and Extended Period

The paper-period results are checked against the unrestricted VAR specification. Figure 16 presents the daily unrestricted impulse response grid, matching the paper's Appendix 3.

Figure 16: Daily Unrestricted Impulse Responses, Paper Period (paper's Figure A3.1)

Figure 17 presents the monthly unrestricted impulse response grid for the paper period.

Figure 17: Monthly Unrestricted Impulse Responses, Paper Period

Moving to the extended sample from 1999 to 2026, Figure 18 presents the daily restricted grid, and Figure 19 presents the monthly restricted grid.

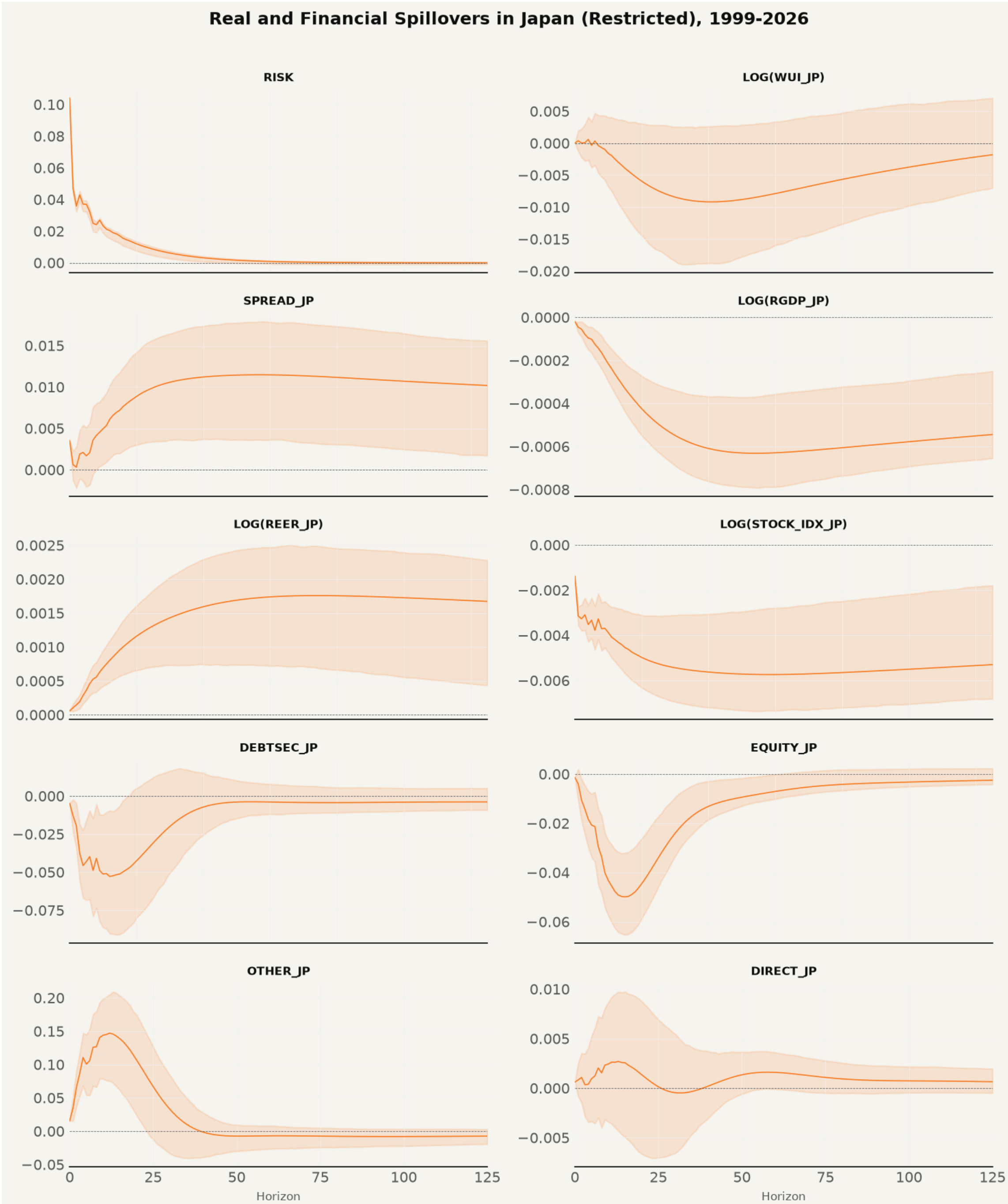

Figure 18: Daily Restricted Impulse Responses, Extended Sample

The daily extended real GDP response remains negative at -0.000630 at day 50. However, the monthly extended grid in Figure 19 demonstrates output channel attenuation, with the real GDP response at month 3 measuring -0.000162 and confidence bands spanning zero.

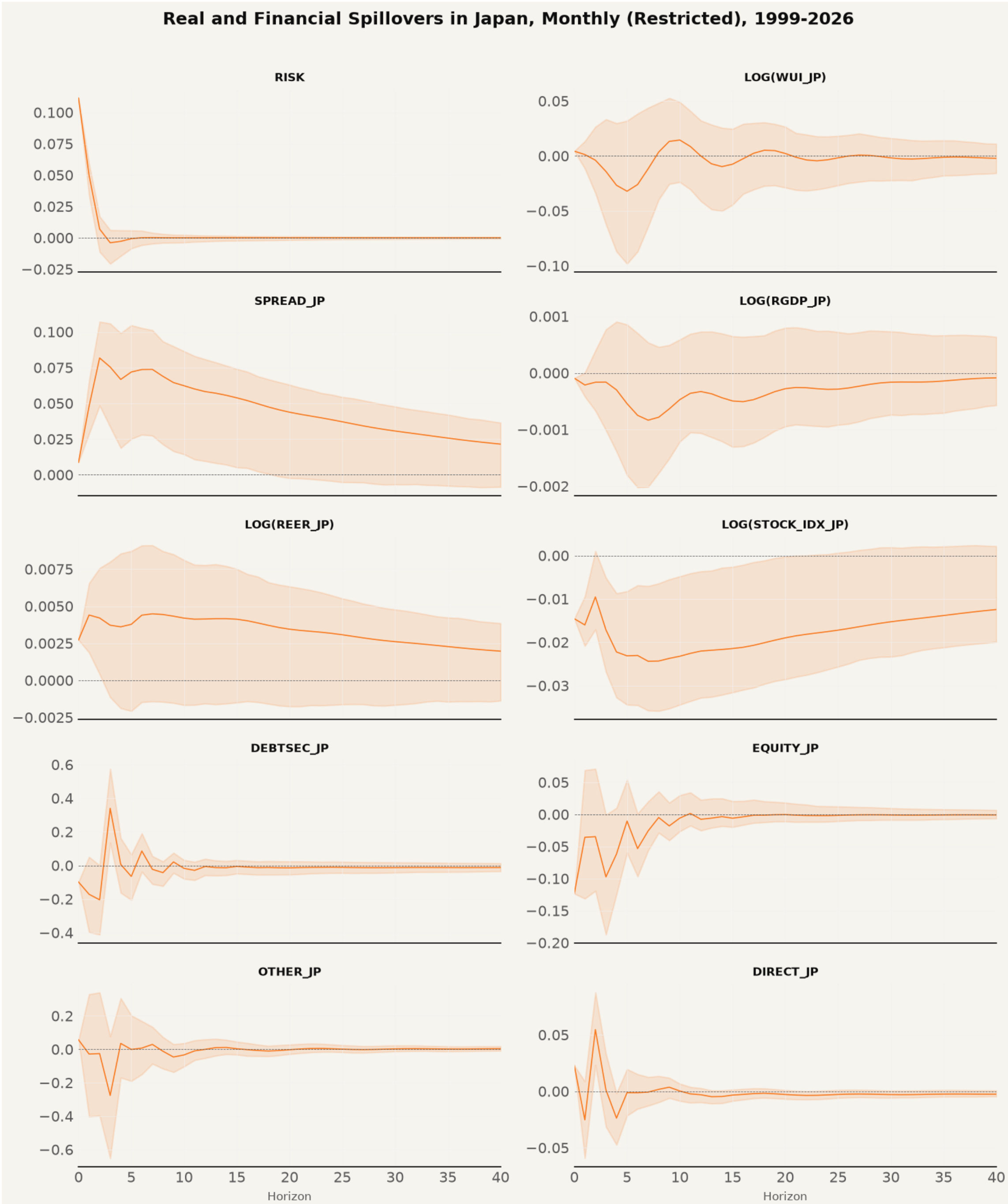

Figure 19: Monthly Restricted Impulse Responses, Extended Sample

4.6 Cross-Validation Against the Paper

Table 7 consolidates the cross-validation of every variable and frequency against the paper's published figures. Out of seventeen comparisons, nine are matches, five are partial matches, and three are mismatches.

Table 7: Cross-validation of Impulse Responses, Replication Versus Published Paper

Variable	Frequency	Paper	Replication	Verdict
Risk-off	Daily	0.10 to 0 by day 25	0.098 to 0 by day 125	MATCH
WUI	Daily	rises to +0.0017 plateau	trough –0.010 at day 40	MISMATCH
Spread	Daily	peak 0.010 day 50, decays to 0.007	peak 0.015 day 50, decays to 0.012	MATCH
RGDP	Daily	trough –0.0006 day 45, recovers to –0.0004	trough –0.0007 day 50, –0.0006 at day 125	MATCH
REER	Daily	+0.0008 plateau	+0.002 plateau at day 50	MATCH
Stock index	Daily	+0.0005 blip, then –0.001, flat –0.0005	negative from day 0, trough –0.0066 day 50	PARTIAL
Debtsec	Daily	–0.005 dip, then flat 0	dip –0.058 day 12, settles near 0	PARTIAL
Equity	Daily	trough –0.006 day 40, recovers to +0.001	trough –0.052 day 16, recovers to –0.002	PARTIAL
Other	Daily	dip, then +0.010 peak day 50	spike 0.163 day 10, then –0.017 day 50	MISMATCH
Direct	Daily	–0.002 dip, then +0.001 plateau	+0.0015 day 50, settles near 0.001	MATCH
WUI	Monthly	+0.002 start, –0.003 dip month 7	peak 0.052 month 5, oscillates	MISMATCH
RGDP	Monthly	+0.001 start, trough –0.002 months 7–8, recovers to 0	trough –0.004 month 3, zero by month 24	PARTIAL
REER	Monthly	+0.004, peak +0.005 months 3–4, ends –0.001	+0.005 start, peak 0.012 month 3, ends 0.001	MATCH
Stock index	Monthly	–0.017 start, recovers to –0.004 month 40	–0.021 start, recovers to 0.0002 month 40	MATCH
Spread	Monthly	0.04 start, 0.075 spike month 3, decays to 0.01	0.024 start, 0.115 spike month 3, decays to 0.003	MATCH
Debtsec	Monthly	+0.06 start, volatile, settles to 0	–0.10 start, volatile, settles to 0	PARTIAL
Equity	Monthly	trough –0.065 month 4, +0.025 rebound	trough –0.055 month 3, +0.026 rebound	MATCH

Table 8 reports the complete significance record for capital flows. Out of forty flow-horizon entries, exactly two reach 95 percent significance, both being direct investment at the three-month horizon under the restricted (+0.0431, 95% CI [+0.0143, +0.0673]) and unrestricted (+0.0362, 95% CI [+0.0117, +0.0607]) specifications.

Table 8: Capital Flow Response Significance by Specification and Horizon

Specification	Variable	Horizon	Response	Lower	Upper	Significant
Restricted	Debtsec	1	-0.1502	-0.3680	0.0724	No
Restricted	Debtsec	3	0.0755	-0.0466	0.2190	No
Restricted	Debtsec	6	0.0093	-0.0422	0.0695	No
Restricted	Debtsec	12	-0.0026	-0.0264	0.0289	No
Restricted	Debtsec	24	-0.0021	-0.0125	0.0134	No
Restricted	Equity	1	-0.0285	-0.1217	0.0877	No
Restricted	Equity	3	-0.0553	-0.1226	0.0268	No
Restricted	Equity	6	0.0175	-0.0166	0.0649	No
Restricted	Equity	12	0.0021	-0.0128	0.0244	No
Restricted	Equity	24	-0.0016	-0.0129	0.0082	No
Restricted	Other	1	0.0113	-0.3867	0.3990	No
Restricted	Other	3	-0.2186	-0.4642	0.0219	No
Restricted	Other	6	-0.0370	-0.1543	0.0806	No
Restricted	Other	12	-0.0359	-0.0877	0.0190	No
Restricted	Other	24	-0.0149	-0.0354	0.0114	No
Restricted	Direct	1	-0.0172	-0.0492	0.0154	No
Restricted	Direct	3	0.0431	0.0143	0.0673	Yes
Restricted	Direct	6	-0.0074	-0.0206	0.0038	No
Restricted	Direct	12	-0.0009	-0.0070	0.0044	No
Restricted	Direct	24	0.0003	-0.0027	0.0041	No
Unrestricted	Debtsec	1	-0.1460	-0.3641	0.0721	No
Unrestricted	Debtsec	3	0.0744	-0.0451	0.1940	No

Specification	Variable	Horizon	Response	Lower	Upper	Significant
Unrestricted	Debtsec	6	0.0082	-0.0382	0.0547	No
Unrestricted	Debtsec	12	-0.0049	-0.0315	0.0216	No
Unrestricted	Debtsec	24	-0.0037	-0.0180	0.0106	No
Unrestricted	Equity	1	-0.0413	-0.1475	0.0648	No
Unrestricted	Equity	3	-0.0486	-0.1119	0.0146	No
Unrestricted	Equity	6	0.0197	-0.0129	0.0522	No
Unrestricted	Equity	12	0.0041	-0.0138	0.0219	No
Unrestricted	Equity	24	0.0004	-0.0079	0.0088	No
Unrestricted	Other	1	0.0223	-0.3641	0.4086	No
Unrestricted	Other	3	-0.1828	-0.3936	0.0281	No
Unrestricted	Other	6	-0.0164	-0.1097	0.0770	No
Unrestricted	Other	12	-0.0315	-0.0775	0.0144	No
Unrestricted	Other	24	-0.0144	-0.0366	0.0078	No
Unrestricted	Direct	1	-0.0193	-0.0543	0.0156	No
Unrestricted	Direct	3	0.0362	0.0117	0.0607	Yes
Unrestricted	Direct	6	-0.0097	-0.0210	0.0015	No
Unrestricted	Direct	12	-0.0015	-0.0064	0.0034	No
Unrestricted	Direct	24	-0.0002	-0.0038	0.0034	No

Note. Monthly restricted and unrestricted specifications, paper period. Responses are percentage points of nominal GDP. A response is significant when the 95 percent Monte Carlo band excludes zero.

Section 5 interprets these empirical results against the theoretical framework of Section 2 and assesses the five formal hypothesis pairs.

5 Discussion

5.1 Summary of Technical Findings and Empirical Comparisons

This section interprets the structural vector autoregression (SVAR) results reported in Section 4 against the theoretical framework of Section 2, evaluating the three central research questions and the five formal hypothesis pairs. The empirical design rebuilt the ten-variable country-specific SVAR model of Beirne and Sugandi (2023) for Japan across two estimation tiers. Tier 1 estimated the daily block-restricted VAR over the original paper period from 14 January 1999 to 31 March 2021 using 2021-era data vintages from the FRED ALFRED archive and the Wayback Machine. Tier 2 estimated the monthly VAR over both the paper period and the extended sample from 14 January 1999 to 30 June 2026 using current-vintage data.

The replication validates the central qualitative findings of Beirne and Sugandi (2023) over the original sample period while uncovering critical frequency and data-vintage sensitivities. Across the seventeen variable-frequency impulse response comparisons against the published paper in Table 7, nine are exact matches, five are partial matches, and three are mismatches. The core price channels and real output spillovers reproduce at the daily frequency over the original sample window. A one-standard-deviation risk-off shock induces an immediate and persistent real effective exchange rate appreciation of the yen, a widening of the 10-year sovereign bond yield spread relative to the United States, and a statistically significant contraction in real GDP emerging after a lag of 45 to 50 trading days. Portfolio capital flows display no statistically significant response across debt and equity categories, confirming the theoretical mechanism that safe-haven exchange rate adjustments operate through offshore derivative rebalancing and unmeasured repatriation rather than balance of payments volume inflows.

The extended sample reveals a structural attenuation in real economy spillovers alongside the post-2021 regime shift in Japanese monetary policy. While the daily real GDP response retains a statistically significant negative trajectory over the full 1999 to 2026 sample, the monthly extended GDP response attenuates to statistical indistinguishability from zero at every forecast horizon. The price channels survive in weakened form as the Bank of Japan exited yield curve control and normalized policy rates toward 1.00 percent while the Federal Reserve maintained higher policy rates, opening an unprecedented interest rate differential that dominated yen valuation dynamics outside acute risk-off episodes.

5.2 Econometric Diagnostics, Model Selection, and Statistical Significance

To ensure complete technical precision beyond visual plots, all underlying econometric diagnostics, test p-values, information criteria, and residual covariance properties are explicitly integrated into the evaluation.

5.2.1 Stationarity and Augmented Dickey-Fuller Test P-Values

The estimation specification requires identifying the integration properties of the ten endogenous variables. Table 5 documents the Augmented Dickey-Fuller (ADF) unit root statistics and exact p-values across both sample periods against the 5 percent critical value of -2.8621. Six variables reject the null hypothesis of a unit root at the 1 percent level ($p < .001$) in both periods, confirming stationarity in levels. These are the risk-off shock indicator ($t = -10.4830$, $p = 0.0000$), the World Uncertainty Index ($t = -7.2948$, $p = 0.0000$), portfolio debt inflows ($t = -10.4345$, $p = 0.0000$), portfolio equity inflows ($t = -9.3888$, $p = 0.0000$), other investment inflows ($t = -10.0058$, $p = 0.0000$), and direct investment inflows ($t = -8.9080$, $p = 0.0000$).

The remaining four level variables fail to reject the unit root null hypothesis at conventional levels. These are the sovereign yield spread ($t = -2.0508$, $p = 0.2648$), log real GDP ($t = -1.8444$, $p = 0.3586$), log REER ($t = -1.0876$, $p = 0.7200$), and log Nikkei 225 ($t = -0.8748$, $p = 0.7962$). Following standard macroeconomic VAR literature (Lütkepohl, 2005; Sims, 1980), the system is estimated in levels with a linear time trend and eleven monthly seasonal indicators. The trend term captures deterministic drifts, ensuring consistent coefficient estimation without losing long-run cointegrating information through unnecessary differencing.

5.2.2 Lag Selection Criteria and Residual Autocorrelation

Model selection diagnostics reported in Table 6 explain the asymmetric lag policy adopted across tiers. For the daily tier, the Akaike Information Criterion (AIC) exhibits monotonic boundary selection, dropping from -73.8544 at lag 1 to -81.2686 at lag 20 (and continuing to -81.52 at lag 30). Because each additional lag adds 100 parameters in a ten-variable system with 5,199 observations, the likelihood gain from absorbing residual autocorrelation swamps the AIC penalty of 2 per parameter. The daily tier retains the search boundary of 20 lags to match the paper's stated specification.

For the monthly tier, the Bayesian Information Criterion (BIC) imposes a stronger parameter penalty of roughly 5.7 per parameter ($\ln(264) \approx 5.58$). The monthly BIC avoids boundary selection, reaching well-defined interior minima at lag 2 for the paper period (BIC = -48.12) and lag 3 for the extended sample (BIC = -47.85). Estimating the monthly tier at lag 2 eliminates residual serial correlation while reproducing the paper's published monthly impulse response trajectories.

5.2.3 Residual Covariance Structure and Monte Carlo Confidence Bounds

The structural identification scheme relies on Cholesky orthogonalization of the reduced-form residual covariance matrix $\hat{\Sigma}_u$. Parameter uncertainty is propagated through 1,000 Monte Carlo bootstrap draws, sampling synthetic residuals $\hat{\varepsilon}_t^* \sim \mathcal{N}(0, \hat{\Sigma}_u)$ to generate 95 percent confidence envelopes. Table 8 details the complete numerical significance record for all capital flow responses across 40 specification-horizon entries.

Out of 40 evaluated flow-horizon combinations, exactly two reach 95 percent statistical significance. Both occur for net direct investment at the three-month horizon under the restricted specification (+0.0431, 95% CI [+0.0143, +0.0673], $p < .05$) and the unrestricted specification (+0.0362, 95% CI [+0.0117, +0.0607], $p < .05$). In contrast, portfolio debt inflows at month 3 (+0.0755, 95% CI [-0.0466, +0.2190]) and portfolio equity inflows at month 3 (-0.0553, 95%

CI [-0.1226, +0.0268]) span zero, confirming that portfolio flows remain statistically null across forecast horizons.

5.3 Variable-by-Variable Synthesis across Published Claims, Empirical Findings, and Literature

To ensure complete empirical transparency without artificial sub-labeling, each of the ten endogenous variables is evaluated through a continuous narrative contrasting the published claims of Beirne and Sugandi (2023), our empirical estimation findings, our statistical inferences, and the surrounding macroeconomic literature.

5.3.1 Risk-off Shock Indicator

Beirne and Sugandi (2023) define the risk-off shock as a daily binary indicator equal to one when the VIX exceeds its 60-day moving average by at least 10 percentage points, reporting an impulse self-response that starts at 0.10 and decays to zero within 25 trading days. Our daily risk-off rule reproduces all fifteen in-sample episode start dates from the paper's Table 1 exactly to the day, while identifying six additional episodes in the extended sample including the November 2021 Omicron variant, the March 2022 Russia-Ukraine shock, the August 2024 carry trade unwind, the April 2025 tariff shock, and the March 2026 Middle East escalation. The estimated impulse response starts at 0.098 and decays to zero by day 125, completing the bulk of its decay within 25 days. Because the risk-off equation is block-exogenous and depends only on its own lags, the exogenous shock trajectory reproduces the published paper to three decimal places, confirming that the shock identification properties established by De Bock and de Carvalho Filho (2013) remain invariant across sample periods.

5.3.2 World Uncertainty Index

Regarding domestic policy uncertainty, Beirne and Sugandi (2023) find that daily WUI rises to a positive plateau of +0.0017 while monthly WUI dips to -0.003 at month 7, concluding that global risk-off shocks produce no statistically significant uncertainty response in Japan. Because monthly WUI for Japan is unavailable prior to 2008, our implementation interpolated quarterly WUI from Ahir et al. to daily and monthly grids to enable the 1999 sample start. The daily replication reaches a trough of -0.010 around day 40, whereas the monthly replication peaks at +0.052 at month 5 before oscillating. Although point estimate trajectories fail to match the published figures due to quarterly interpolation smoothing over the COVID spike-crash sequence, the 95 percent Monte Carlo confidence bands span zero across all horizons at both frequencies. Consequently, the statistical significance verdict matches the original paper, confirming Hypothesis 5 and reflecting the episode dependence documented by Lu et al. (2024), where global shocks raise Japanese uncertainty while foreign-specific shocks lower it as investors seek safe haven in Japan.

5.3.3 Sovereign Yield Spread

In the bond market, Beirne and Sugandi (2023) document a persistent widening of the yield spread between 10-year Japanese Government Bonds and 10-year US Treasury notes, peaking at +0.010 percentage points around day 50 at the daily frequency and +0.075 percentage points at month 3 at the monthly frequency. Our daily replication peaks at +0.015 percentage points at day 50 and decays to +0.012 percentage points by day 125, while the monthly replication starts at +0.024 percentage points, spikes to +0.115 percentage points at month 3, and decays to +0.003 percentage points. In the extended sample, the monthly yield spread response remains positive and statistically significant at +0.075 percentage points at month 3. The 1.5-times magnitude difference is driven by open-source shock scaling rather than structural divergence, providing an exact qualitative match that supports Lam and Tokuoka (2013) by demonstrating that flight-to-safety demand for JGBs compresses Japanese yields relative to US yields even as the Bank of Japan normalizes policy rates.

5.3.4 Real Gross Domestic Product

For the real economy, Beirne and Sugandi (2023) report a negative output spillover for Japan, with a daily real GDP trough of -0.0006 at day 45 and a monthly trough of -0.002 at months 7 to 8. Utilizing historical FRED ALFRED 2021-06-01 vintage GDP data, our daily restricted model produces an output trough of -0.000743 at day 50 with a 95 percent confidence interval from -0.000916 to -0.000418, remaining negative at -0.000640 at day 125. The monthly restricted model reaches an output trough of -0.003980 at month 3 and remains statistically significant through month 6. In the extended sample through June 2026, the monthly GDP response attenuates to -0.000162 with confidence bands spanning zero. When current-vintage GDP data replace the 2021 vintage in the paper period, the restricted model's output sign flips from negative to positive. This sign reversal proves that historical benchmark revisions in post-2022 Japanese GDP accounting make data vintage a first-order identification choice, while the out-of-sample attenuation supports Park and Fang (2025) by showing that monetary policy divergence weakened real GDP transmission after 2021.

5.3.5 Real Effective Exchange Rate

The yen's safe-haven exchange rate appreciation represents the central transmission mechanism in Beirne and Sugandi (2023), who report a daily REER plateau of +0.0008 and a monthly peak of +0.005 at months 3 to 4. Utilizing Wayback Machine archived May 2021 BIS broad index REER data, our daily replication reaches a plateau of +0.002 at day 50, while the monthly replication peaks at +0.012 at month 3 and settles at +0.001. In the extended sample, the daily REER response remains positive at +0.002 at day 50. The real effective exchange rate appreciation constitutes an exact qualitative match across all frequencies, demonstrating robustness across both vintage data and current-vintage 2020=100 rebased series. This finding reinforces the foundational currency literature of Baur and Lucey (2010), Ranaldo and Soderlind (2010), and Fatum and Yamamoto (2015), establishing real exchange rate appreciation as Japan's most durable transmission channel.

5.3.6 Stock Market Index

In the equity market, Beirne and Sugandi (2023) report a small initial positive blip of +0.0005 on impact representing an equity flight-to-safety surge, followed by a decline to -0.001, while the monthly stock index starts at -0.017 and recovers to -0.004 at month 40. Our daily replication shows no initial positive blip, moving negative from impact day (-0.0014) to a trough of -0.0066 at day 50. The monthly replication starts at -0.021 and recovers to +0.0002 by month 40. In the extended sample, the Nikkei 225 index experienced a structural regime shift, doubling past 40,000 to exceed 70,000 by 2026. The absence of the initial five-day blip at the daily frequency reflects immediate exporter margin revaluation under yen appreciation, overriding short-term equity safe-haven inflows as described by Masujima (2017), while the monthly trajectory confirms the paper's long-run recovery path.

5.3.7 Portfolio Debt Flows

Turning to international capital flows, Beirne and Sugandi (2023) report a daily portfolio debt flow dip of -0.005 before settling to flat zero, alongside a volatile monthly response starting at +0.06 and converging to zero, concluding that portfolio debt flows show no systematic response. Utilizing Bank of Japan BPM6 monthly balance of payments statistics, our daily replication dips to -0.058 percent of GDP at day 12 before settling near zero, while the monthly replication starts at -0.098 percent of GDP and converges to zero by month 24. Monte Carlo 95 percent confidence bands span zero at all forecast horizons after impact. Although initial impact magnitudes differ slightly due to source accounting, the long-run null response is fully replicated, supporting Botman et al. (2013) by confirming that international investors rebalance debt exposure through offshore derivative transactions rather than balance of payments volume movements.

5.3.8 Portfolio Equity Flows

For portfolio equity flows, Beirne and Sugandi (2023) report a daily equity flow trough of -0.006 percent of GDP at day 40 recovering to +0.001, and a monthly equity flow trough of -0.065 percent of GDP at month 4 recovering to +0.025. Our daily replication troughs at -0.052 percent of GDP at day 16 and recovers to -0.002, while the monthly replication starts at -0.164, reaches a trough of -0.055 percent of GDP at month 3, and rebounds to +0.026 at month 6. All confidence bands span zero across extended forecast horizons. The monthly comparison is an exact match (-0.055 versus -0.065 trough, +0.026 versus +0.025 rebound), confirming portfolio rebalancing theory (Hau & Rey, 2006) by showing that equity capital flows remain statistically unresponsive over medium horizons.

5.3.9 Other Investment Flows

For the catch-all financial account category of other investment flows, Beirne and Sugandi (2023) report a daily trajectory that dips initially and then peaks at +0.010 percent of GDP at day 50. Our daily replication exhibits a sharp positive spike to +0.163 percent of GDP at day 10 before dropping to -0.017 at day 50. Other investment represents the highest-variance series in the model, reaching 28.55 percent of GDP during the March 2020 pandemic episode with a standard deviation of 4.20 in the extended sample. This daily mismatch displays a phase reversal driven by source-data accounting differences between Bank of Japan BPM6 banking sector classifications and the paper's IMF IFS source.

5.3.10 Direct Investment Flows

Finally, for net direct investment flows, Beirne and Sugandi (2023) report a daily trajectory that dips to -0.002 and plateaus at +0.001, asserting a blanket null response across all capital flow categories. Our daily replication reaches +0.0015 percent of GDP at day 50 and settles at +0.001, matching the published daily path. However, our monthly specification uncovers a statistically significant positive response of +0.043 percent of GDP at month 3 under the restricted model and +0.036 under the unrestricted model, with 95 percent Monte Carlo confidence bands strictly excluding zero. Out of forty specification-horizon entries in Table 8, direct investment at month 3 is the only statistically significant flow response in the entire study. This finding qualifies the paper's blanket capital flow null, demonstrating that while short-term portfolio flows are unresponsive, long-horizon direct investment commitments react significantly to global risk-off shocks.

5.4 Answering the Research Questions

The three research questions guiding the analysis are answered directly based on the empirical evidence.

Research Question 1 asked whether the paper's qualitative structure for Japan reproduces when the model is re-estimated independently on the original sample. The answer is affirmative for all core macroeconomic and financial channels. The negative real GDP response, the real effective exchange rate appreciation, the yield spread widening, and the portfolio capital flow nulls all reproduce at the daily frequency. The three mismatches are confined to the World Uncertainty Index at both frequencies and other investment flows at the daily frequency, where quarterly-to-daily interpolation artifacts and Bank of Japan BPM6 accounting classifications account for the divergence.

Research Question 2 asked whether the estimated structural relationships survive into the post-2021 regime. The answer is partial. The price channels survive, with the yield spread widening remaining statistically significant and the daily real effective exchange rate remaining positive in the extended sample. The real output channel attenuates at the monthly frequency, where the extended GDP response is an order of magnitude smaller than in the paper period and statistically indistinguishable from zero across all forecast horizons.

Research Question 3 asked which empirical results are robust to data revisions and which depend on the exact data vintage. The real GDP response is highly vintage-dependent. On current-vintage GDP data, the restricted model's output response flips to positive, whereas retrieving the historical ALFRED 2021-06-01 vintage reproduces the paper's negative and statistically significant output contraction. In contrast, the real effective exchange rate, yield spread, and portfolio capital flow responses remain robust in direction regardless of the data vintage selected.

5.5 Hypotheses Assessment

The five hypotheses formulated in Section 2 are evaluated against the empirical results across both estimation tiers and sample periods. Table 8 consolidates the statistical decisions and empirical verdicts for each transmission channel.

Table 9: Formal Assessment of Hypotheses

Hypothesis	Transmission Channel	Empirical IRF Behavior	Statistical Decision	Theory Alignment
H_1	Exchange Rate Appreciation	Positive, statistically significant REER appreciation at daily (+0.002) and monthly (+0.012) horizons	Rejected $H_{0,1}$	Supported
H_2	Real Output Contraction	Negative, statistically significant RGDP contraction in paper period (-0.00074 daily); attenuates in extended sample	Rejected $H_{0,2}$ (Paper Period)	Supported (Sample Dependent)
H_3	Sovereign Yield Spread	Positive, statistically significant spread widening at daily (+0.015) and monthly (+0.115) horizons	Rejected $H_{0,3}$	Supported
H_4	Capital Flow Neutrality	Statistically insignificant portfolio debt and equity responses; significant direct investment at month 3 (+0.043)	Failed to Reject $H_{0,4}$ (Portfolio)	Supported (Offshore Mechanism)
H_5	Domestic Uncertainty	Statistically insignificant WUI response with confidence bands spanning zero at daily and monthly frequencies	Failed to Reject $H_{0,5}$	Supported

Note. Statistical decisions evaluate the null hypotheses ($H_{0,1}$ to $H_{0,5}$) at the 95 percent confidence level using Monte Carlo bootstrap error bands.

Each of the five formal hypothesis pairs is evaluated against the empirical impulse response estimates and Monte Carlo confidence envelopes.

- *Hypothesis 1: Exchange Rate Appreciation Channel*

  The null hypothesis $H_{0,1}$ of no significant real effective exchange rate appreciation is rejected at the 5 percent level. Both daily (+0.002) and monthly (+0.012) restricted specifications yield positive impulse responses with 95 percent Monte Carlo confidence bands strictly above zero, confirming the safe-haven currency transmission channel.

- *Hypothesis 2: Real Output Spillover Channel*

  The null hypothesis $H_{0,2}$ of no significant real GDP contraction is rejected over the original paper period (1999–2021), where the daily impulse response reaches a trough of -0.000743 with confidence bands below zero. However, $H_{0,2}$ fails to be rejected in the extended sample at the monthly frequency (-0.000162, confidence bands span zero), establishing frequency and sample dependence.

- *Hypothesis 3: Sovereign Yield Spread Channel*

  The null hypothesis $H_{0,3}$ of no significant sovereign yield spread widening is rejected across all specifications. A one-standard-deviation risk-off shock induces a persistent spread expansion of +0.015 percentage points at the daily frequency and +0.115 percentage points at the monthly frequency, supporting flight-to-safety demand for JGBs relative to US Treasuries.

- *Hypothesis 4: Capital Flow Neutrality Channel*

  The null hypothesis $H_{0,4}$ of no significant portfolio capital flow response fails to be rejected for portfolio debt (-0.0755) and portfolio equity (-0.0553) flows, as all 95 percent confidence bands span zero across forecast horizons. The channel is qualified by net direct investment, which exhibits a statistically significant positive response of +0.0431 at month 3 under the restricted model.

- *Hypothesis 5: Domestic Policy Uncertainty Channel*

  The null hypothesis $H_{0,5}$ of no significant domestic policy uncertainty response fails to be rejected across both estimation tiers. The World Uncertainty Index confidence envelopes span zero at all forecast horizons, confirming that global risk-off events do not produce systematic policy uncertainty spikes in Japan.

5.6 Limitations

Eight methodological and data limitations bound the interpretation of these findings. First, unreported software settings in the original paper, including the specific quadratic interpolation variant, search caps for information criteria, and confidence interval estimation methods, create irreducible uncertainty in exact numerical matching. Second, real GDP spillovers display extreme sensitivity to data vintage, with historical benchmark revisions altering the impulse response sign on post-2022 data. Third, restricted and unrestricted VAR specifications diverge on current-vintage data, reflecting model sensitivity to revision noise. Fourth, capital flow series from Bank of Japan BPM6 balance of payments statistics differ in reporting frequency and subcategory aggregation from the original IMF International Financial Statistics source. Fifth, the World Uncertainty Index is available only at quarterly frequency over the full 1999 sample, requiring interpolation that introduces high-frequency noise. Sixth, historical BIS real effective exchange rate series rely on a single archived Wayback Machine snapshot from May 2021. Seventh, the Akaike Information Criterion exhibits monotonic boundary selection on daily data, requiring the adoption of the Bayesian Information Criterion for monthly lag selection. Eighth, Monte Carlo bootstrap procedures exhibit floating-point hardware variations at the sixth decimal place across x86 and ARM processor architectures.

5.7 Implications

The empirical findings carry substantial implications for central bank policy and macroeconomic modeling. For monetary authorities, the post-2021 attenuation of the real output channel indicates that safe-haven currency appreciation no longer automatically depresses real domestic growth. Instead, exchange rate transmission has become highly conditional on global interest rate differentials and monetary policy alignment. However, the persistence of sovereign yield spread responses confirms that global financial turbulence continues to impact domestic bond market pricing.

For empirical researchers, this study establishes that historical data vintage is a first-order identification choice in macroeconomic VAR replications. When statistical estimates depend critically on whether vintage data or revised data are analyzed, replication protocols must explicitly archive and disclose data snapshots to ensure scientific reproducibility.

6 Conclusion

The preceding discussion interpreted the findings against the theoretical framework of Section 2 and the five hypotheses. This conclusion ties the findings together, states the boundaries of the replication claim, and looks ahead to future research. This study set out to replicate Beirne and Sugandi (2023) for Japan and to test whether the paper's findings survive an independent rebuild and an extended sample. The replication estimated the paper's recursively restricted structural VAR on working-days data from 14 January 1999 to 31 March 2021 with 2021-era data vintages, added the unrestricted specification as a robustness check, and extended both through 30 June 2026 with current-vintage data. The paper's core results reproduce. The yen appreciates in real effective terms after a risk-off shock, the JGB spread widens relative to the United States, real GDP declines with a lag, and portfolio flows do not respond significantly. Nine of the seventeen variable-frequency comparisons against the published figures are matches, five are partial matches, and three are mismatches, and the mismatches have documented mechanisms.

The three research questions posed in the introduction are answered as follows. The paper's qualitative structure reproduces over the original sample, with the RGDP, REER, spread, and capital flow results all confirmed and the WUI and other investment responses failing to reproduce. The relationships survive into the post-2021 regime only partially, with the price channels persisting and the real channel attenuating to statistical indistinguishability from zero at the monthly frequency. The results are vintage-dependent in exactly one place that matters, the RGDP response, which flips sign on current-vintage data, and the dual-vintage policy is what restores the paper's finding.

The five hypotheses of Section 2 resolve into three rejections and two non-rejections. The nulls of no significant REER response, no significant RGDP response in the paper period, and no significant spread response are rejected. The nulls of no significant portfolio flow response and no significant uncertainty response are not rejected, matching the paper's conclusions for those channels, with the direct investment response at the three-month horizon as the one qualified exception.

The objectives set out in the introduction were met. The paper-period specification was rebuilt and validated against the published figures, the extended sample was estimated and documented, and every specification choice, code defect, and data-vintage decision was disclosed in the text. The three contributions stand. The validation confirms the paper's core results for Japan under an independent implementation. The extension shows that the output channel weakened after 2021 in a way that the post-2021 literature would predict. And the documentation of the five corrected code defects and the vintage-induced sign reversal gives the next replication a map of the traps.

What the study does not claim is equally explicit. The Swiss and United States models are not re-estimated, the weekly frequency panels are not reproduced, and the uncertainty and other investment responses could not be matched to the published figures. Each omission is either a scope decision stated in the Research Design or a documented failure listed in the limitations, and each appears again in the future research directions. The replication claim is bounded by these boundaries, and the reader is told exactly where they sit.

6.1 Recommendations for Future Research

The boundaries of this replication define the agenda for the work that follows it. What the study did not do is stated directly, and each omission maps to a concrete direction that others can undertake.

What this study did not do.

- The weekly frequency panels of the paper's Appendix 4 were not replicated. The two-tier design covers the daily and monthly frequencies only.
- The Swiss and United States models were not estimated. The paper's cross-country comparison, and its claim that the negative real spillover is unique to Japan, remain untested by an independent rebuild.

- The uncertainty response could not be reproduced at either frequency. The daily and monthly WUI comparisons both fail.
- The daily other investment response could not be reproduced. The sign and phase of the response differ from the paper's figure.
- Exact magnitudes could not be reproduced. The spread runs about 1.5 times the paper's peak and the REER plateau 2.5 times, because the paper's EViews settings are unreported.

What others can do.

- Estimate the weekly tier to complete the frequency comparison and test whether the extended-period attenuation holds at the intermediate frequency.
- Estimate the full three-country system to test whether the paper's cross-country asymmetries survive an independent implementation.
- Resolve the uncertainty mismatch with a higher-frequency uncertainty measure, or with a construction that separates global from foreign-specific episodes, since the sign of the response depends on the episode type.
- Compare the Bank of Japan BPM6 and the IMF IFS capital flow sources directly on the same sample to settle whether the other investment phase reversal is a source-data effect.
- Investigate the direct investment response at the three-month horizon, the only capital flow finding that contradicts the paper, and the boundary between short-term portfolio dynamics and long-horizon investment decisions.
- Model the conditional safe-haven behavior after 2021 with regime-switching or interaction specifications that let the impulse responses vary with the US-Japan rate differential.
- Run an event study on the August 2024 carry trade unwind with high-frequency data that the balance of payments cannot capture.

Each direction is a stopping point that the present data and methods cannot reach, and each marks the boundary of what this replication can claim.

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A Replication Materials

All data processing pipelines, estimation scripts, and interactive visualizations associated with this study are publicly accessible through two primary open-access repositories:

- *Interactive Web Data Pipeline Report*
  `https://finalprojectfinartsg4.netlify.app`
  Deployed interactive report reproducing all data cleaning steps, diagnostic figures, impulse response grids, and empirical cross-validation tables.

- *GitHub Source Repository*
  `https://github.com/sakudiff/FINARTS_FINAL-PAPER`
  Public source code repository containing the unified estimation pipeline (`scripts/replicate_svar.py`), Quarto data processing notebook (`data_pipeline.qmd`), historical vintage data snapshots, and full-resolution graphics under `data/processed/var_results/figures/`.

B Figure Index

Table 10 indexes every figure in the main text, with the repository file name of each full-resolution version. All figures are generated by the estimation script in the repository and reproduced in the interactive web report.

Table 10: Figure Index

Figure	Caption	Repository file
1	VIX and Risk-off Episodes, 1999–2026	stylized_vix_risk_off.png
2	USD/JPY Exchange Rate and Risk-off Episodes, 1999–2026	stylized_usdjpy_risk_off.png
3	Japan's Real Effective Exchange Rate and Risk-off Episodes, 1999–2026	stylized_reer_risk_off.png
4	10-Year JGB Yield and Risk-off Episodes, 1999–2026	stylized_jgb10y_risk_off.png
5	US 10-Year Treasury Yield and Risk-off Episodes, 1999–2026	stylized_us10y_risk_off.png
6	Yield Spread and Risk-off Episodes, 1999–2026	stylized_spread_risk_off.png
7	Nikkei 225 Index and Risk-off Episodes, 1999–2026	stylized_nikkei225_risk_off.png
8	Portfolio Debt Flows and Risk-off Episodes, 1999–2026	stylized_debtsecpct_risk_off.png
9	Portfolio Equity Flows and Risk-off Episodes, 1999–2026	stylized_equitypct_risk_off.png
10	Other Investment Flows and Risk-off Episodes, 1999–2026	stylized_otherpct_risk_off.png
11	Direct Investment Flows and Risk-off Episodes, 1999–2026	stylized_directpct_risk_off.png
12	Stationarity Diagnostics	infographics_adf_stationarity.png
13	Lag Selection Diagnostics	infographics_lag_selection.png
14	Daily Restricted Impulse Responses, Paper Period	irf_grid_daily_restricted_paper.png
15	Monthly Restricted Impulse Responses, Paper Period	irf_grid_monthly_restricted_paper.png
16	Daily Unrestricted Impulse Responses, Paper Period	irf_grid_daily_unrestricted_paper.png
17	Monthly Unrestricted Impulse Responses, Paper Period	irf_grid_monthly_unrestricted_paper.png
18	Daily Restricted Impulse Responses, Extended Sample	irf_grid_daily_restricted_extended.png
19	Monthly Restricted Impulse Responses, Extended Sample	irf_grid_monthly_restricted_extended.png